

SCIENCE

INSTANT LESSONS POWERPOINT

20
sets of
PowerPoint
slides

THE SKILLS OF SCIENCE

SCIENTIFIC METHOD

BOOK 2

Donna Bennett



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This book was commissioned by Jennifer Gough, project management was by Piper Print Media, Suellen Entwisle designed the cover and the PowerPoint presentations, and David Dickson supplied the illustrations.



Donna Bennett is a highly qualified and experienced teacher. She is the winner of two national awards, the CRA Mining Company Fellowship and the BHP Science Teacher Award, for innovative and creative delivery of science. She is the Director of Science Education Australia and her science workshop programs are always popular with teachers and students alike. Donna has written several successful science education books, established the Queensland Science Marathon and has taken her 'Science for Bush Kids' program to regional areas. She has travelled internationally where her special form of science education has been in demand.

Donna believes that science is the greatest adventure that has ever been devised. She sees teachers as consultants and cheerleaders who provide probing questions, enthusiasm and confidence and who tease out an understanding of the basic notions of science. Teachers are a source of knowledge and information, which is shared in order to guide student exploration, rather than define it. As an author she has a clear and effective style and believes that interactive science investigations contribute most to learning when they are part of a well-thought out plan that triggers the natural curiosity of students. She provides a freedom that allows students the pleasure of building their own knowledge and understanding. Once they get over the shock of having this type of responsibility, and experience the thrill of succeeding, nothing can stop them.

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The scientific method is simple and makes really good sense. Some people make a lifetime career out of defining scientific method – others just use it!

Science is like a game about understanding the universe and this game is played by special rules called the 'scientific method'. The game usually involves asking a question, gathering information about the question, forming the hypothesis, putting it to the test and telling others about what you found out. When people play this game, they must remember not only that the rules are flexible and provide only a general plan, but that it usually leaves out the important bits. Anyone who loves science knows that these missing bits, such as hunches, wrong turns, intuition and the ideas that didn't work make scientific discovery the most challenging, exhilarating, frustrating, yet exciting adventure ever devised.

This series is about the nature of science and testing ideas. It encourages an everyday perception of science, conceptual understanding and thinking skills.

The activities:

- encourage students to learn how to think and how to explore scientifically
- encourage the ability to see a good question, investigate it and draw evidence-based conclusions
- introduce the notion of a fair comparison as a precursor to the idea of 'controlled experiments'
- emphasise that in science, nothing can be proved – only disproved!
- will instill a confidence in conversing about science and questioning claims about scientific matters
- develop the ability to make informed decisions about global and everyday issues.
- separate science from myth, considered opinion from hasty judgment and will help students to realise that of all the things that are scientifically possible, not all are desirable.

It is worth remembering what Galileo Galilei (1546-1642), the 'father' of scientific method said: *The experiment is the ordeal through which all ideas must pass before they are accepted as truth.* The game of science is about curiosity, which turns into the satisfaction of knowledge – it should begin with some fun and end with wisdom.

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What does it include?

The topics in this title are complemented by the information and activities contained on the CD-ROM at the back of this book.

The CD-ROM includes:

- 20 PowerPoint presentations to accompany each of the 20 topics in this book, which summarise and extend the content.
- 20 follow-up activity sheets, each based on the matching PowerPoint presentation. These are supplied as PDF documents.

Three video-clips are also provided on the CD-ROM, as extension material for the topics listed below.

Topic 2 – The hierarchy of classification, using the horse and the zebra as examples.

Topic 19 – a: Exploring the issues of genetic modification

b: How genetic modification occurs

Note to the teacher

To get the most out of these teaching resources, teachers are encouraged to preview the Powerpoint presentations and accompanying activity sheets before using them in class.

Using the CD-ROM

In the Windows environment, the CD-ROM will autorun. From the Main menu, simply click the PowerPoint presentation you wish to view.

In the Macintosh environment, double-click the file index.pps. From the Main menu, click the PowerPoint presentation you wish to view.

You can also open each presentation from within the PowerPoint application.

The Slide Show view is used to display each PowerPoint presentation on the screen. It is recommended that you use an external projector for whole-class use.

To access the activity sheets, simply click the pencil icon on the appropriate topic.

To access the video-clips, navigate through Windows Explorer (for Windows users) or Finder (for Macintosh users) to the CD-ROM, and open the folder 'Videos'. Double-click the video-clip you wish to play.

Technical specifications

Windows

- The CD-ROM will autorun in the Windows environment, with an operating system of Windows 97 or later.
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a version prior to Office 2000, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Macintosh

- Double-click the file index.pps on the CD-ROM. From the Main menu, click on the presentation you wish to view.
- To view a fully featured presentation, and edit it, you must have the PowerPoint 2004 application installed. (i.e. Office 2004 for Macintosh or later)
- To edit a presentation, open it from within the PowerPoint application. Note that if editing these PowerPoint presentations in a Macintosh environment, cosmetic losses will occur in some animations, and visual effects such as drop shadows may be lost. These losses are cosmetic only, and will not affect the clarity and content of the slide show.

Zoom in!

Our five senses are a ‘radar’ system, feeding us information about the environment around us. Each sense supplies the brain with messages and the brain tells the body how to respond. All of this information from ears, eyes, nose, skin and tongue makes us curious. When we are curious, we ask questions. Scientists use all of their senses to gather information and find answers to their questions.

Activity 1

1.1 Draw lines to match each sense with the correct job.

SENSE	JOB
• eye	• detects chemicals in substances in the mouth
• nose	• detects temperature, touch, pain
• tongue	• detects vibrations in the air
• ear	• detects light
• skin	• detects chemicals in the air

1.2 Which sense or senses are being used in each situation below?

SITUATION	SENSE/SENSES USED
a Using a microscope	
b Recording the croaks made by selected tree frogs	
c Sorting a rock collection according to colour and smoothness	

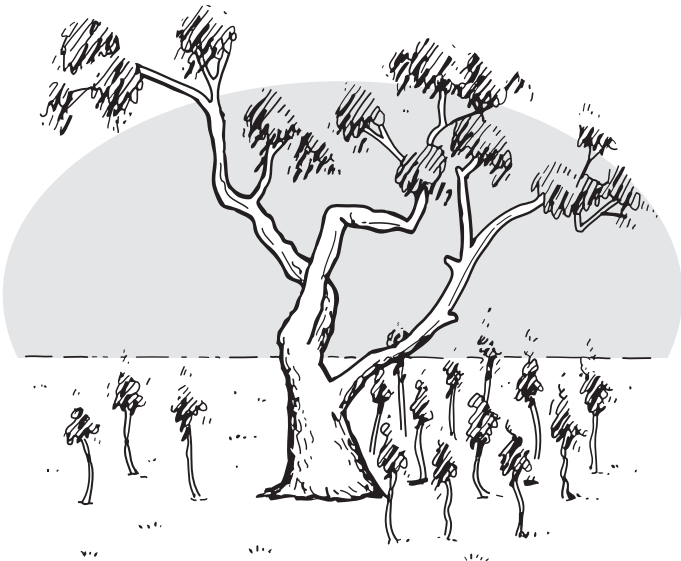
Is observing the same as seeing?

Observing is an important skill in science. In science, we depend mostly on our sense of sight, but observation is far more than just seeing. Observing means working with all of your senses as well as your brain to put information together.

Activity 2

2.1 A student looked at the picture at the right and saw a tree with seedlings growing under it.

- a What do you see?
- b Do you notice anything else about where the seedlings are growing?
- c What further questions could be raised by this observation?



Can you believe everything you observe?

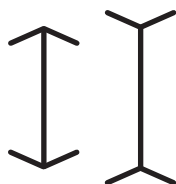
We cannot always trust our sense organs. Our observations can be influenced by many things. What a person sees can depend on what they expect to see. Different people notice different things when looking at the same scene.

Imagine the scene of a bushfire. A person running away from the fire, a child watching the fire and a firefighter would all make different observations about the scene.

Activity 3

3.1 Look at the optical illusions below. What do you see when you look at each one?

What do you see in the picture at the right?



Which vertical line is longer?

Is the hat taller than it is wide?



Which horizontal line is longer?

Can we rely on our sense organs for accuracy?

Accurate measurement is important in science. Using measurements means that observations are more likely to be understood in the same way by other people. For example, a person living in Hobart and a person living in Darwin probably have different ideas of what 'hot weather' is. However, both people would have the same understanding of the statement: *It was 30 degrees outside.*

We can help our sense organs to measure accurately by using instruments and equipment.

Activity 4

4.1 For each pair of observations, decide which one is more likely to be understood in the same way by other people – the first or the second.

a ▲ It was a cold day.

▲ The temperature fell to 6 degrees Celsius at 4.00 pm.

b ▲ The band was very noisy.

▲ From the front seat, the sound from the band measured 110 decibels.

4.2 Using measurement instruments helps to make more accurate observations. Which sense does each of these measurement instruments help?

a ruler

b stopwatch

c thermometer

What happens after observation?

In science, we can take observations one step further and use them to make **inferences**. Sometimes it is not easy to tell the difference between an observation and an inference.

Consider this situation: Bella and Lee went fishing at sunset, using worms for bait.

An observation is something we are certain about. For example, Bella's observation afterwards was: *Every time we threw a fishing line into the water we caught a fish.* We have direct evidence that this statement is true. An inference is an explanation or deduction based on observed data. For example, Lee's inference was: *It's easy to catch fish at sunset.* Bella's inference was: *It's easy to catch fish when you use worms for bait.* The girls made different assumptions about their fishing success.

Activity 5

5.1 A student recorded the information below while watching a candle burn. Which statement is an observation?

a The flame is yellow.

b The wick is coated with carbon.

c The wax burns.

5.2 A scientist observed that chemical A mixed with water is safe. She also observed that chemical B mixed with water is safe. From these two observations, can she conclude that mixing chemicals A and B with water is safe? Explain your answer.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The process of classification

To classify things is to arrange them into groups. Classification is done by humans for the convenience of humans. It helps people to order the world around them, and therefore improve their communication about it and their understanding of it. Classification involves observation, comparison, description, naming and developing sub-groups.

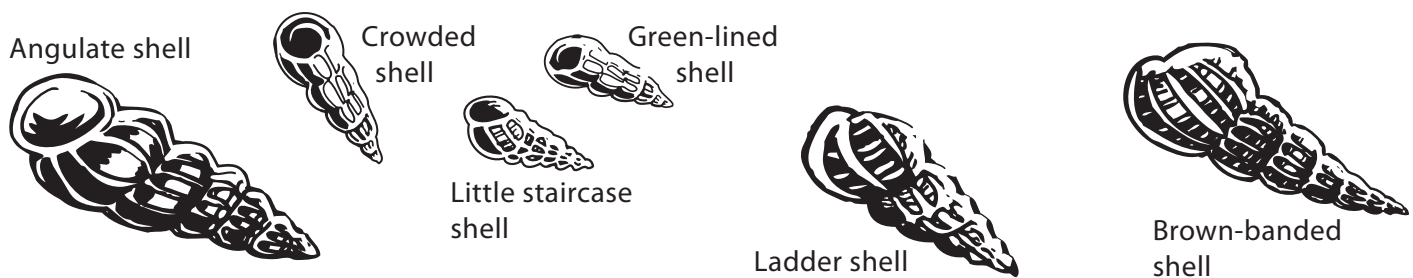
When classifying, observe objects carefully and work out how one is different from another. Remember that small details can be very important and some differences can be more important than others.

Activity 1

- 1.1 Maia’s teacher provided a variety of stationery items for the class, and asked the students to classify them. Maia drew up the table below and has classified most of the objects.
- a Complete the task by writing these items in the correct places in the table: *paperclip, pencil, notepad, marker pen, exercise book, eraser*.
- | WRITING | | | ORGANISING |
|---------------------------|--------------------------|----------------------|----------------------------------|
| For writing with | For writing on | For deleting writing | For connecting |
| • ballpoint pen
•
• | • sticky notes
•
• | • white-out
• | • bulldog clip
• staples
• |
- b Describe the similarities and differences between each group Maia chose.
- c What other ways could Maia have classified the stationery items?

Activity 2

2.1 Shells of the type shown below are usually white and polished, but some are coloured.



- a Study the shells carefully and note any differences between them. Read the names of the shells for clues.
- b Using the table below, organise the shells into two groups using size as the characteristic. How many shells did you place in the group of shells more than 3 cm long?

Common name of shell	Less than 3 cm long	More than 3 cm long	Coloured	White
Angulate shell				
Crowded shell				
Little staircase shell				
Green-lined shell				
Ladder shell				
Brown-banded shell				

- c Study the names of the shells and any shading in the illustration. Then classify the shells again, using colour as the characteristic. How many shells did you place in the group ‘Coloured’?

- d

You have classified these shells using two different characteristics – size and then colour. Are the groupings the same in both classification systems?
- e

Which do you think would be the most useful for classification – classifying according to size, colour or both? Why?
- 2.2

Crocodiles and snakes belong to a large group of animals called reptiles. However, they do not belong to the same sub-group. In the table below, tick the characteristics which apply to reptiles. Then tick the characteristics that divide crocodiles and snakes into the two sub-groups. The first row has been completed for you.

	Characteristic	Reptiles	Crocodile sub-group	Snake sub-group
a	Hinged jaws	✓	✓	
b	Tough, scaly, water-proof skin			
c	Water-dwelling			
d	Eggs with soft, waterproof shell			
e	Legless			
f	Has a skeleton			

A name is just a name ... Or is it?

When a life form is classified, biologists describe it and give it a scientific name. This name identifies it and shows which group it belongs to in the living world.

Names are often based on the unique characteristics of the life form or where it grows. For instance, the earliest flower fossil ever discovered originated under water and was found in China. Scientists called it *Archaeofructus sinensis*, which is Latin for 'ancient fruit from China'.

Scientists have agreed to use Latin as the international language for identifying and naming living things. Scientific names have two parts: genus and species. The genus has a capital letter and the species does not, and the whole name is written in italics. For example, *Micropus rufus* is the scientific name for the red kangaroo.

The adoption of this binomial classification system for living things by the scientific community is credited to Carolus Linnaeus, who lived in the 18th century.

Activity 3

3.1 The scientific name for the venus flytrap is *Dionaea muscipula*.

- a

What is the common name?
- b

Which is the species name?
- c

Which is the genus name?
- d

What do we call this system of naming?
- To find the scientific name of an organism you do not know, look for the things that make it unique – such as unusual colours, strange markings and distinctive shapes. For example, an egg is oval in shape, white in colour, hard on the outside and mushy on the inside! Once its characteristics are described, then it is easy to classify it into the correct group and to use an identification guide to find the scientific name for the organism – unless it is a completely new organism, of course!

Activity 4

4.1 A scientist looking at a venus flytrap described the plant as: 'a large, purplish and green plant with big teeth and a ravenous appetite for flies'.

- a

Which details in the description are the most important clues for identifying this plant?
- b

What makes this plant unique?
- c

Which feature of this plant, as described above, is not entirely accurate?
- d

Into which group of plants would you place this plant – ferns, orchids, carnivorous plants or grasses?
- 

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Another way to solve a problem is to see it differently.

Scientific discoveries often occur because someone has been able to see a less obvious solution to a problem. Sometimes people put unnecessary restrictions on the options that are available.

Activity 1

- 1.1 Imagine six drinking glasses standing on a table in a row, numbered in order from 1 to 6. Numbers 1, 2 and 6 are empty. Numbers 3, 4, and 5 are full of water.
 - a By moving or touching only one glass, how could you rearrange the order so that the glasses alternate between full and empty – that is, no two empty glasses are beside each other, and no two full glasses are beside each other? *Tip: Draw the six glasses.*
 - b What restriction did you have to overcome before you could solve this puzzle?
- 1.2 Suppose you have a drawer holding six red socks and six green socks.
 - a If you had your eyes closed, how many socks would you need to take out before you could be sure of having at least one matching pair?
 - b What restriction did you have to overcome before you could solve this puzzle?

Imagination

Science is about solving puzzles, which needs not only imagination, but also determination.

Activity 2

- 2.1 A good imagination is a real asset for a scientist. Close your eyes and imagine each of these things in turn.
 - a the smell of bananas
 - b the sound of a cricket bat hitting a ball
 - c what it would be like to follow the path of a sound wave
- 2.2 When news of Alexander Graham Bell's new invention, the telephone, reached Britain, the chief engineer's response was: *The Americans have need of the telephone, but we do not. We have plenty of messenger boys.* Why is his statement an example of a 'failure of imagination'?
- 2.3 At the end of the nineteenth century, carriage builders of the time refused to believe that the invention of the motor car would have any impact on the horse and carriage trade. How would this lack of imagination affect the carriage builder?
- 2.4 Five hundred years ago, Leonardo da Vinci envisaged a world with cars, tanks, flying machines and multi-level bridges.
 - a List five things you imagine will be possible in 40 years' time.
 - b Now list five things you think will probably be unchanged in 40 years' time.

Reverse thinking

It may be useful to try reverse thinking when solving a problem. In other words, think of the strategies that you could use, then reverse them.

Activity 3

- 3.1 Irving Langmuir improved the electric light bulb by using reverse thinking. The problem was that residual oxygen inside the bulb kept causing the filament to burn. Instead of trying to remove the traces of oxygen from inside the bulb, he pumped a different gas into the bulb – a gas that would not support combustion.
 - a State the problem Langmuir was trying to solve.
 - b How did Langmuir apply reverse thinking in order to solve this problem?
 - c What group of gases must Langmuir have used?

Mind experiments – dreaming up solutions or scientific breakthrough?

Scientists can often visualise meaning, ideas and problems, rather than achieving all of their scientific discoveries by carrying out practical experiments.

Activity 4

- 4.1** When was the last time that you figured out an everyday problem by using only your mind to work out a solution? What was the problem and what was the solution?
- 4.2** Scientists sometimes just see things and know things without being able to explain how they know. Has this ever happened to you? What was it that you knew?
- 4.3** Margaret Knight was an inventor who lived in the 1800s. When she was 12, she worked in a cotton mill. There was an accident caused by a snagged thread which made a spindle fly off a machine, injuring a worker. Margaret invented a covered shuttle to prevent workers from being injured. Margaret went on to invent many more things, and registered 26 patents.
 - a** List three situations where accidents occur.
 - b** For each situation you listed, suggest two inventions that could reduce the chances of accidents happening.

Scientists do everything they do with a creative twist.

Some problems or puzzles need an entirely new approach or a whole new way of thinking.

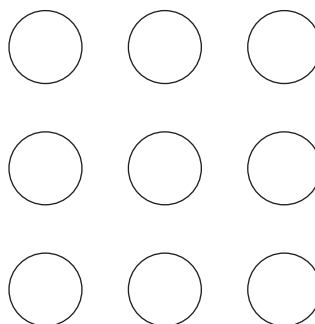
Creativity is not a process that you have power over. It tends to work away inside your mind and then unexpectedly pops out of your subconscious when you least expect it. It also tends to disappear altogether just when you need it most!

It may not be possible to turn you into a Curie, Knight, Einstein or da Vinci, but let's try a little lateral thinking. The knack with lateral thinking is to allow your mind to wander over, under and around the problem without making any real effort to find the solution.

Activity 5

- 5.1** Join these nine circles *without* lifting your pen.

- a** How many times did you have to change direction?
- b** Join the circles again, but this time change direction only four times.
- c** Repeat, changing direction only three times.
- d** Can you join the circles changing direction only twice?



Clue – Can you see the pattern of circles in a way other than a square?

- 5.2** Consider the following list of people's surnames:

Salk, Crick, Ingenhousz, Einstein, Newton, Copernicus, Edison, Gilcrest.

- a** Who is the odd person out? Why?
- b** Referring to the names again, give another reason why this person is the odd one out.
- 5.3** A woman who has no map is walking along a track in a strange country. She comes to an intersection where the signpost has been knocked down. There is no one to ask for directions – how can she find her way?
- 5.4** What two things are unique about the number 854 917 632?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

What is research?

Research is an activity that is based on the work of others. It is the process through which new knowledge is discovered. The word *research* means ‘to investigate thoroughly’. There are different forms of research.

Your task

You are the Research Technician who is part of the Research and Development (R & D) team of the **Fast Foot** athletic shoe company. You have been approached by two professional athletes, requesting a design for new athletic shoes which will give them the edge over their competition. The characteristics they want developed are foot support and durability, combined with flexibility and spring.

Your plan is to research materials that may be suitable for the redevelopment of the soles of current shoe designs. You will design these shoes especially for these athletes, but will also be applying your research to the manufacture of new season basketball, tennis, running and cross-trainer shoes for your company. To make the research profitable, you will also be selling these ‘designer’ sports shoes to the general public. The shoes must therefore not only perform at an exceptional standard, but also look good.



Activity 1

1.1 As the Research Technician, you are in charge of testing each new polymer formulation for specific characteristics, and ensuring a quality product. You have carried out five experiments on three new polymer formulations that could be suitable for the new shoes. The results are summarised in the ratings table below.

TEST	POLYMER A RATING	POLYMER B RATING	POLYMER C RATING
Texture and consistency (softness)	High	Medium	Low
Resistance to flow (viscosity)	Low (flows easily)	Medium	High (resists flow)
Impact recovery (flexibility)	High	Medium	Low
Strength	Low	Medium	High
Elasticity (bounce)	High	Medium	Medium

- a** Describe polymer A.
- b** Describe polymer C.
- c** Your company conducts tests on its products and on the materials that are used. This is called ‘quality control’, and is standard practice throughout industry. Why is this type of research or testing necessary?

Research publishing

This is a system through which researchers make their work available for a wider audience. Most research is published as journal articles. Scientists present their work in the journals and publications that are relevant to their field of work. These publications are accepted as valid contributions to human understanding.

Activity 2

2.1 Read the following research report about the biology of foot movement and the impact on athletic shoes.

RESEARCH REPORT

As an athlete rolls forward onto the forefoot, the force exerted may be between four and seven times an athlete’s body weight. The muscles spread evenly across the bones of the forefoot, absorbing the energy that propels the foot off the ground for the next stride. During foot fall, the body weight strikes on the heel, shifts forward over the midfoot and rolls forward onto the toes.

If shoes do not absorb the energy, the ankles, shins and knees must take the force. Athletes will pay for shoes that will not only enhance their performance but help them to avoid injury.

Different parts of a shoe demand different requirements from a shoe:

- ▲ The ‘midsole’ is responsible for protecting the ball of the foot from shock as the impact of the heel rolls forwards. Soft, fluid plastic that is flexible and has good elasticity or bounce is suitable.
- ▲ The ‘foresole’ is the part that pushes off from the ground, so plastic must absorb as little energy as possible. The athlete must be able to ‘spring’ away from the ground. A suitable plastic would be flexible but also offer good shock absorption.
- ▲ The ‘rearsole’ takes the repeated abuse of each footfall. It hits the ground first with up to eight times the weight of the athlete impacting on the heel of the foot. A good shock absorber deforms when struck but returns quickly ready for the next impact and provides the least amount of bounce.

- a Which polymer from the table in activity 1 will provide an energetic advantage and therefore a performance advantage upon push off?
- b Where in the sole of a shoe would you use a viscous plastic?
- c Where in the sole of a shoe would you want a polymer with the fastest flow rate?
- d Which polymer would you use if you needed to combine some elastic recovery with some bounce?
- 2.2 Complete the table below to help you design a shoe sole that provides support, flexibility, durability and spring.

POLYMER	FORESOLE	MIDSOLE	REARSOLE	COMMENTS
A				
B				
C				

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Scientific enquiry

Taking a scientific approach to answering a question can be rather like being a detective in a mystery story. The story usually starts with a problem; let's say somebody is dead. The detective usually guesses what is most likely to have happened – devises a hypothesis. What was the cause of death – murder? How was it committed – gunshot, poisoning? The detective then tries to explain what happened – conducts the experiment. She does this by looking for such things as motive, opportunity and murder weapon. These things all provide evidence to support her hypothesis.

Testing a hypothesis

A good hypothesis is one that can be tested by experimentation. Consider this question:

Do the pupils in our eyes respond to light and dark?

It can be turned into this hypothesis:

Dim light will cause the pupils in our eyes to dilate more than bright light.

This statement can now be tested by experimenting.

Activity 1

1.1 Which hypothesis about mouldy sandwiches can be tested satisfactorily?

- a Low temperatures slow the growth of bread mould.
- b Bread mould doesn't like being in the fridge.

1.2 Write each question about bushfires as a hypothesis.

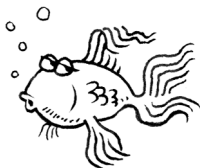
- a Does the sound of trees burning travel further at night than in the day?
- b Do native animals smell smoke sooner than humans?
- c Does burning grass produce more heat than burning trees?

Writing a hypothesis

In science, asking questions leads to formulating a hypothesis. A hypothesis is an educated or plausible guess, based on a question. A hypothesis can be devised by using a combination of common sense and previous experience, and useful hypotheses can be written in different ways.

Activity 2

2.1 Your gold fish seems to breathe more slowly after its tank is relocated to a cooler place. This observation is based on how an organism behaves when something is altered. Use the box below to complete a suitable hypothesis for testing this observation.



The _____ will _____ when _____

↑
In this space, describe the behaviour you are interested in.

↑
In this space, describe what you think will happen.

↑
In this space, describe the thing that you want to alter.

2.2 You are on safari in the African savannah and notice that the tour operator knows exactly where to find the lions. In the middle of the day, they are mostly found sprawled on the branches of umbrella thorn trees. When you are studying animals in the field you cannot change or manipulate the conditions. Write a hypothesis to test this observation.



are

In this space, describe the type of organism that you are interested in investigating.

In this space, make a statement about its distribution, size, density, etc.

2.3 The table below shows observations about whales recorded over a number of years.

WHALE SPECIES	LIVE IN GROUPS	LENGTH	BODY SHAPE	BREACHING RATE
A	no	45 m	thin	rarely
B	no	35 m	thin	rarely
C	yes	16 m	round	often
D	yes	17 m	round	often
E	yes	32 m	round	often
F	yes	15 m	round	often

- a Whales breach in order to communicate. Which of the observations below best supports this hypothesis?
 - ▲ Long whales breach less often.
 - ▲ Thin whales breach less often than round whales.
 - ▲ Breaching whales flick their tails when they re-enter the water.
 - ▲ Whales that breach most often live in groups.
- b Which of the following hypotheses is also supported by the observations above?
 - ▲ Round whales need more air than thin whales.
 - ▲ Whales breach so that they can dive more deeply.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Experimenting with variables

A **hypothesis** proposes a possible cause and effect relationship between two variables. This involves selecting one variable and suggesting how this variable might influence a different variable. The variable selected as the possible cause is called the **independent variable**. Because the investigator can regulate this variable, it is sometimes also referred to as the manipulated variable. The variable that changes as a result of alterations to the independent variable is called the responding or **dependent variable**.

For example, consider the hypothesis: *Salt dissolves faster in hot water than in cold water.*

In the investigation to test this hypothesis, the independent variable is the temperature of the water and the dependent variable is the dissolving speed of the salt.

Confidence with a single variable

Often experiments need to be conducted on a large group of people or a large group of items; for example, in testing medications or in the quality testing of new products.

Where there is very little variation between individuals in a group to be tested, usually the group is divided into two equal parts and it is assumed that the parts are identical. This approach is used when we can be reasonably sure that all the variables can be controlled.

Remember the basic rule of investigation – change one thing, measure one thing and keep everything else the same.

Activity 1

- 1.1 A manufacturer is making sunglasses out of a new ‘unbreakable’ material. The sunglasses are being marketed as ‘KrakFree Sunglasses’. To test this claim, 500 pairs of the new sunglasses have been dropped onto many different surfaces.
- a What hypothesis did the manufacturer test?
 - b What was the independent variable?
 - c What was the dependant variable?
 - d What variables did they have to control?

Controlled experiments

Controlled experiments are a form of experimental procedure used when it is difficult to control precisely all of the variables in an experiment. This procedure is often used when living organisms are involved, as many variables will be related to the actual organisms; for example, age, sex and genotype. Controlled experiments involve conducting two similar experiments side by side. Only one of these experiments tests the independent variable. This allows comparisons to be made between the responses of both groups. In this way, the responses that are the result of changes in the independent variable can be identified as they should only occur in the experimental group and not in the control group.

EXPERIMENTAL GROUP	CONTROL GROUP
• Must be maintained under normal conditions in all aspects except for the one variable being tested. • Tests the effect of the dependent variable on the independent variable.	• This group is assumed to be identical to the experimental group and must be controlled under normal conditions. • The independent variable is either absent or is kept at a set level and is not allowed to change. • The control group is the basis for comparison of all other observations.

Activity 2

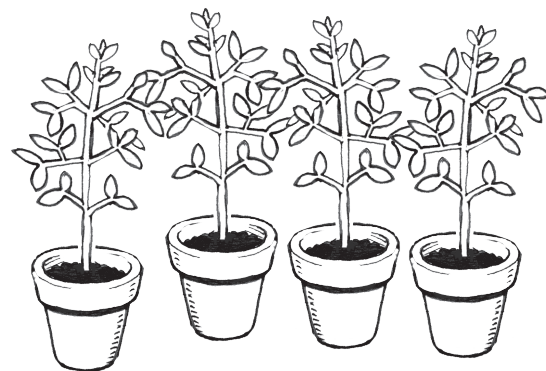
Consider this hypothesis: *At a particular location, the absence of nitrates in the soil is retarding plant growth.*

2.1 In the experiment to test this hypothesis:

- a** what is the independent variable?
- b** what is the dependent variable?

2.2 To test this hypothesis, plant growth in pots where nitrate fertiliser had been applied will be compared with plant growth in pots where nitrate fertiliser had not been applied.

- a** Describe the control group.
- b** Describe the experimental group.
- c** Why is the control group important?



2.3 As well as the level of nitrates in the soil, what other variables may affect plant growth?

2.4 What variables must be controlled for the duration of the experiment?

2.5 Since the plant growth in both the control and experimental groups will be compared, what factors must be controlled so that we can be sure that any difference in growth is the result of the application of fertiliser?

2.6 If the hypothesis is accepted, then what result must we have observed?

2.7 What will we do if the growth of the control group was the same or more than that of the experimental group?

Correlation

Sometimes it is impossible to control all the variables in an experiment, and the best we can do is identify **connections** between variables and the probability of a certain outcome. A correlation therefore suggests that two variables might be linked, but does not provide evidence that they are connected. For example, height and weight are related – taller people are generally heavier than shorter people. This connection is called a correlation, but it does not indicate a cause and effect relationship, such as the one investigated in activity two between plant growth and the application of nitrate fertiliser. A key thing to remember when working with a correlation is that you should never assume that a change in one variable causes a change in another variable.



Positive and negative correlation

If two variables are said to be positively correlated, they increase or decrease together; for example, smoking and lung disease are positively correlated. If two variables are negatively correlated, one goes up while the other goes down; that is, high values of one variable are associated with low values of a second variable. Examples of negative correlations include those between amount of exercise and heart failure and between absence from school and school achievement.

Activity 3

3.1 Do you think the correlation between each pair of variables is positive or negative?

- a** Fat consumption and heart disease
- b** Carbon dioxide levels in the atmosphere and global warming
- c** The quality of chocolate and its price
- d** Time spent watching TV and marks achieved at school



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

How do you design a simple experiment?

Scientists plan and organise experiments in such a way that they learn something from them. Good experiments do not have to be complicated. A straightforward hypothesis will give a clear, unquestionable answer when the experiment has been carried out.

Alcohol and accidents

Read the following information about a scientific trial designed to examine factors that may contribute to traffic accidents.

One factor examined in this trial was people’s alcohol consumption in relation to their reaction times. Two groups of volunteers were tested at the same time and both groups were informed of the purpose of the exercise.

GROUP A	GROUP B
Group A was made up of 20 male university students from the business faculty. The subjects in Group A were given three ‘standard’ drinks containing a total of 40 mL of alcohol and their reaction times were tested.	Group B was of a similar age range to Group A, and made up of 10 female and 6 male students from the veterinary faculty. Group B was given a placebo* in the form of an equivalent amount of a non-alcoholic drink and their reaction times were tested.

** A placebo is a medication that performs no physiological function but may affect the patient psychologically. A placebo system is often used in drug trials. The people in the trials usually do not know if they are receiving the active substance being tested, or the placebo.*

When the results from both groups were averaged, it was found that there was no appreciable difference between them. However, it was noted that the females, on average, had faster reaction times than the males and that older students tended to have slightly faster reaction times than younger students.

Experiments usually consist of a test and a control

The **control set-up** is the set-up where the behaviour is predictable. For example, bread mould in warm, moist conditions grows very well. The control set-up is used as a comparison with other set-ups where the conditions have been varied.

The **experimental set-up** is identical to the control set-up, except that one of the conditions (variables) has been changed. If two or more variables were altered at once, it would be impossible to know which of the variables caused the observed result.

Activity 1

Refer to the **Alcohol and accidents** information above to complete this activity.

1.1 Use the number codes below to answer the following questions.

1	Experimental group	5	Reaction time
2	Dependent variable	6	Group B
3	Alcohol consumption	7	Control group
4	Group A	8	Independent variable

- a Which number corresponds to the independent variable?
- b Which number corresponds to Group A?
- c Which of the following pairs of numbers is a correct combination?
- 1 and 6

• 7 and 6

• 2 and 3

• 5 and 8

- 1.2** Which hypothesis is most likely to be the one the investigator originally set out to test?
- a** Females are less likely to be involved in traffic accidents than males.
 - b** Alcohol slows down people's reaction times.
 - c** Younger people have faster reaction times than older people.
 - d** Business students have a higher tolerance to alcohol than veterinary students.
- 1.3** Which of the following variables do you think is least likely to account for differences in the reaction times between the different individuals in the test?
- a** age
 - b** weather conditions
 - c** sex
 - d** drinking habits
- 1.4** Which of the following variables do you think was most adequately controlled in this investigation?
- a** age
 - b** size of the group
 - c** drinking habits
 - d** sex

Experimental rigour

The stricter you are in setting up your experiment, the more you can rely on the results. Try to think of every possible criticism of your experiment, and modify your experiment to avoid the criticisms. For example, in the bread mould experiment, if the experimental set-up is kept inside a fridge, the control set-up should also be inside a fridge, in case the fridge itself, air-tightness or some other unknown factor affects the growth rate of the mould.

Activity 2

Refer to the **Alcohol and accidents** information on the previous page to complete this activity.

- 2.1** Which of the following **would not** improve the design of this investigation?
- a** Repeat the test over a longer period of time.
 - b** Increase the number of individuals over a longer time.
 - c** Test individual reaction times before and after consumption of the liquid.
 - d** Have university students only in Group A, not in Group B.
- 2.2** What conclusion could best be drawn from this investigation?
- a** University students are better drivers than people not attending university.
 - b** Females have a higher tolerance to alcohol than males.
 - c** Alcohol consumption does not affect a driver's reaction times.
 - d** The investigation was inconclusive because of inadequate experimental design.
- 2.3** Which of the following represents the correct sequence of procedures in a scientific investigation?
- a** prediction ➡ define problem ➡ experiment
 - b** hypothesis ➡ experiment ➡ prediction
 - c** define problem ➡ hypothesis ➡ experiment
 - d** experiment ➡ prediction ➡ conclusion
- 2.4** Which of the following affects the outcome of a properly conducted experiment?
- a** controlled variables
 - b** the control
 - c** the dependent variable
 - d** the independent variable



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Measuring tools

Results of experiments should be measured. Descriptions or opinions are too vague.

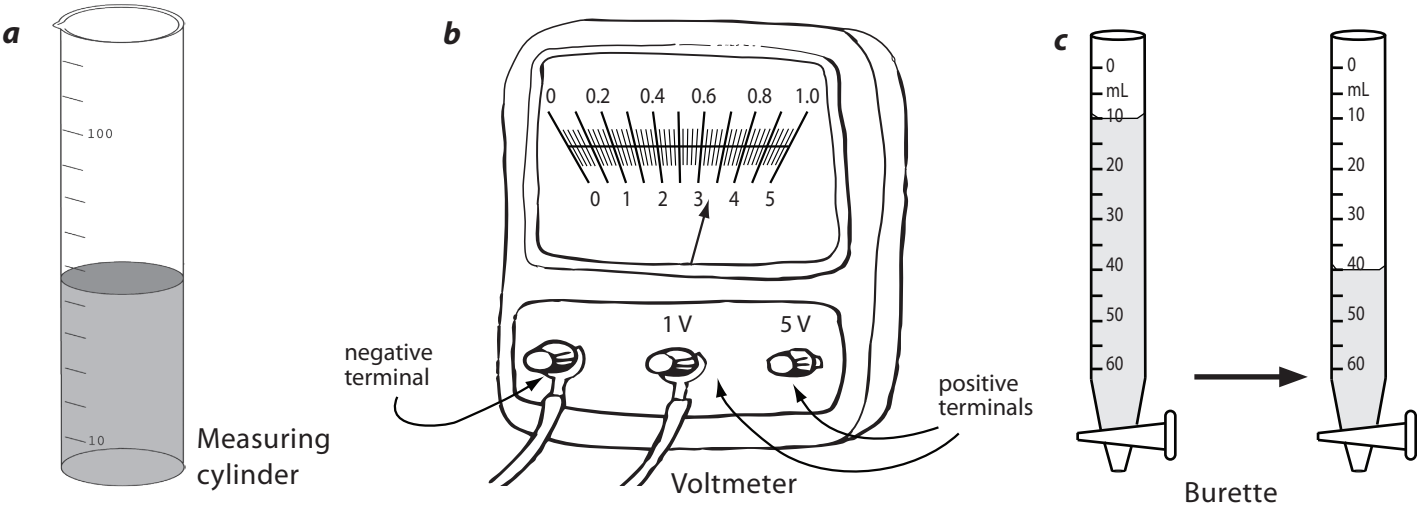
Remember to measure and record data accurately no matter what instrument you are using. It can also be useful to repeat an experiment to check the data.

Precision

Accurate measurements are needed in many everyday activities, like cooking and cabinet making. For scientific experiments it is especially important to be as precise as possible.

Activity 1

1.1 Read the scale on each diagram as accurately as you can and record the readings



1.2 In diagram **b**, the scale to be read depends on which positive terminal is connected. What would the reading be if the other positive terminal was connected?

1.3 Diagram **c** shows a burette adding acid to a flask. How much acid was added?

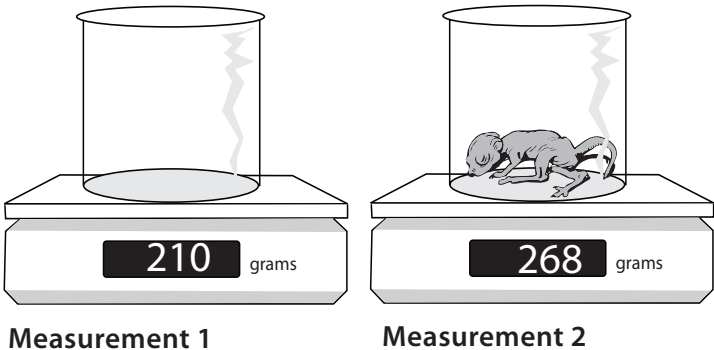
Mass as a tool

The **mass** of an object is the measure of how much matter it contains. Mass is measured in kilograms (kg) or grams (g). To find the mass of an object, you can use simple balancing scales to compare the unknown mass with a known mass.

Activity 2

2.1 A wildlife carer collected data about the mass of a growing wallaby, shown in the table below. The diagram shows the measurements that were made on day 6. Use these measurements to complete the table.

AGE OF THE WALLABY (days)	MASS OF THE WALLABY (grams)	INCREASE IN MASS (grams)
1	46	-
2	47	1
3	49	2
4	51	2
5	54	3
6		



Volume as a tool

The **volume** of an object is the measurement of the amount of space it occupies. It is measured in cubic metres (m^3) or cubic centimetres (cm^3).

Activity 3

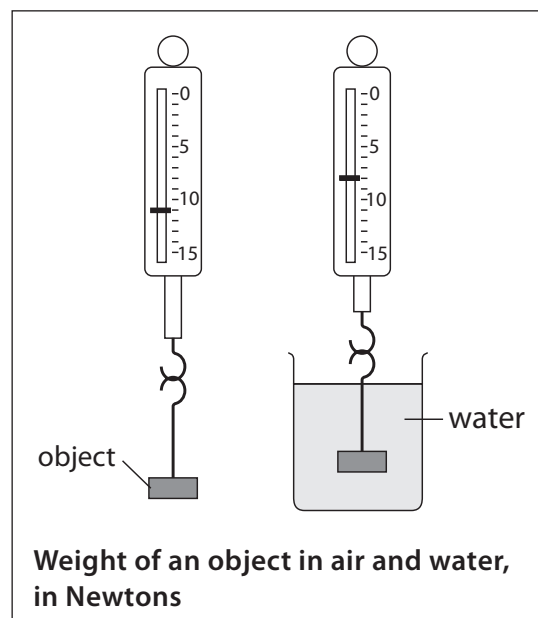
3.1 The volume of regular shaped solids can be found by mathematical calculation.

- a** What is the volume of a block of cheese which measures 4 cm by 10 cm by 2 cm?
- b** What units did you use to express your answer?

3.2 The volume of an irregular shaped solid can be measured by submerging it in water and measuring the amount of water it displaces – this is called displacement. If you dived into a swimming pool and displaced 300 litres of water, what would be your volume?

3.3 The relative density of an object can be measured by comparing the weight of the object in air with the weight of the same object in water. Study the diagram at the right.

- a** What is the name of the instrument being used to measure the weight of the object?
- b** What units are being used?
- c** What is the weight of the object in air?
- d** What is the weight of the object in water?



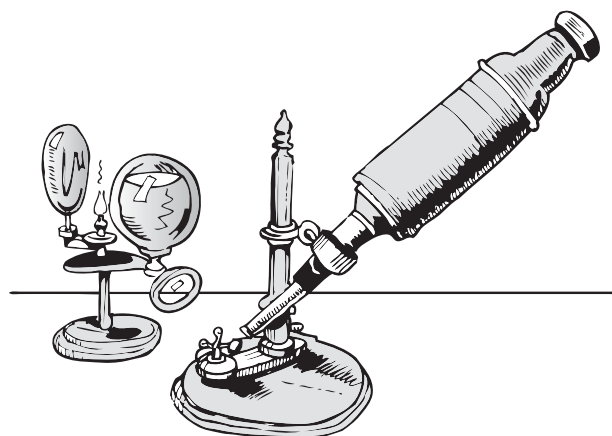
Magnification as a tool

Until the invention of the microscope, scientific study was limited to the examination and comparison of readily visible physical features.

Activity 4

4.1 Write the objects in order according to their ability to magnify, from lowest ability to greatest ability: *light microscope, human eye, electron microscope, magnifying glass.*

4.2 A light microscope built by Robert Hooke in about 1670 is shown at the right. Hooke used a microscope to examine an extremely thin film of dried cork from an oak tree. He gave the name 'cells' to the tiny structures that he saw.



- a** Why did Hooke use such a thin sheet of cork?
- b** The electron microscope is capable of much higher magnification than a light microscope. State one advantage (other than magnification) and one disadvantage of the electron microscope in comparison with the light microscope.

4.3 Write each term beside the correct statement: *resolution, field of view, magnification.*

- a** The amount by which the image is enlarged.
- b** The sharpness at which the specimen can be seen.
- c** The amount of illuminated view seen through a microscope.

4.4 If the field of view was 2.4 mm and the organism was 150 microns in size, how many organisms would fit across the diameter of the field of view?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Accuracy in recording

One basic idea that scientists rely on is that people conducting an experiment will be open, accurate and accountable for their results. Keeping detailed and accurate records is an essential part of this. It helps scientists in recording and reporting on the experimental results, providing accurate evidence of how effective the procedure is and assisting others with reproducing the experimental results.

Presenting measurements

In a laboratory, measurements are rarely jotted down on any piece of paper that is handy. Measurements are usually recorded clearly in a table that is specifically designed for a particular experiment. This contributes to the data being accurate, comprehensive and easy for others to understand.

Activity 1

1.1 Table 1 at the right shows some measurements recorded about performance racing cars. Refer to this table to answer the following questions.

- a How many variables are listed?
- b What is the dependent variable?
- c What units have been used to record time and weight?

1.2 Table 2 shows the same measurements as in Table 1, but does not correctly display the data. List the four major mistakes made in Table 2.

Table 2

Time	Weight
20	0.5
40	1.0
60	1.5

Table 1 – Variation in time to reach 200 km/hour with car weight

Car weight (tonnes)	Time to reach a speed of 200 km/hour (seconds)
0.5	20
1.0	40
1.5	60



Using codes to sample materials

Sampling is a process used by scientists for collecting portions of a material so that they can test the portion to obtain information about the whole material. A material can be animal, vegetable or mineral as well as a solid, a liquid or a gas. Sometimes getting the sample is dangerous or very difficult.

Activity 2

2.1 Consider the codes for types of samples listed below. In the two tables below, write the correct code for each sampling situation and each sampling difficulty.

- ▲ Chemical – C
- ▲ Geological – G
- ▲ Environmental – E
- ▲ Biological – B
- ▲ Food – F

Table 1

CODE (SS)	SAMPLING SITUATION
	Contaminated soil
	Peanut butter
	Air inside a suspect mine shaft
	Horse urine
	Natural gas

Table 2

CODE (SD)	SAMPLING DIFFICULTIES
	Unwilling equine donor
	Prevention of contamination with air
	Affected by a mixture of organic matter and moisture
	Making sure that the food poisoning bugs were actually in the butter that was sampled
	Someone needs to go into the location and not be overcome by poisonous gases

Indicators and tests

In many experiments and biological explorations, different substances have to be identified. There are special tests which help scientists to do this by distinguishing between properties of substances. Many of these tests use indicators, which are substances that ‘indicate’ the presence of other, specific substances.

Activity 3

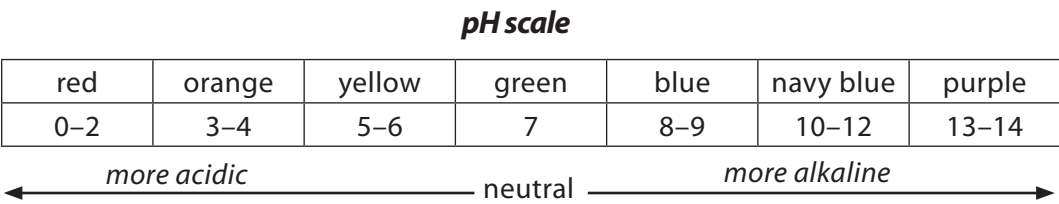
3.1 Consider the six tests listed in the table. Then for each pair of substances below, use the clues to help you decide which test would be best to use to distinguish one substance from the other.

TEST	RESULT
Observation	Use your sight – colour, texture, etc. Use your sense – never taste, touch or smell an unidentified substance.
Water test	Does it sink, float or dissolve?
Iodine test	Iodine turns purple if starch is present
Vinegar test	Substance contains bicarbonate if it bubbles
Acid/base test	Red-ish = acid, blue-ish = basic
Sugar test	Produces bubbles when mixed with yeast and warm water

- CLUES**
- ▲ cream of tartar is an acid
 - ▲ flour and sugar are neutral
 - ▲ flour and talcum powder are the only substances in this activity that do not dissolve in water
 - ▲ table salt and bicarbonate of soda are basic
 - ▲ bicarbonate of soda produces carbon dioxide when mixed with an acid

- a bicarbonate of soda and cream of tartar
- b artificial sweetener and raw sugar
- c chalk dust and cornflour
- d powdered laundry detergent and plain flour
- e instant mashed potatoes and white sand
- f talcum powder and powdered milk

3.2 Indicators are dyes which change colour at different pH values. Universal Indicator can be used to identify acids and bases, and its colour changes at different pH values are shown below.



Each cleaning product in the table below was tested using Universal Indicator, and the resulting colour noted. Identify each one as an acid or a base.

Cleaning product	Colour	Acid or base?
Toothpaste	blue-ish	
Bleach	dark blue	
Shampoo	pale yellow	
Dish detergent	yellowy green	
Lemon juice	orange	



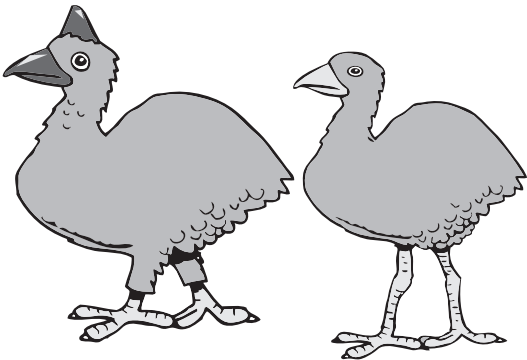
Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Put your boots on!

Most experiments are carried out in a laboratory, but some need to be conducted outside ‘in the field’. The advantage of field experiments is that the observations are conducted in a natural setting rather than in a contrived laboratory environment. However, in field experiments, conditions cannot be controlled with the same precision and certainty as they can in a laboratory.

Activity 1

1.1 In January 2005, a group of scientists carried out research on a rare, ground-dwelling bird that lives in a national park. The drawings at the right are of the rare, ground-dwelling bird under study and a similar, commonly observed ground-dwelling bird that also lives in the park. The drawings have been reduced to 25% of the actual bird size.



Rare, ground-dwelling bird

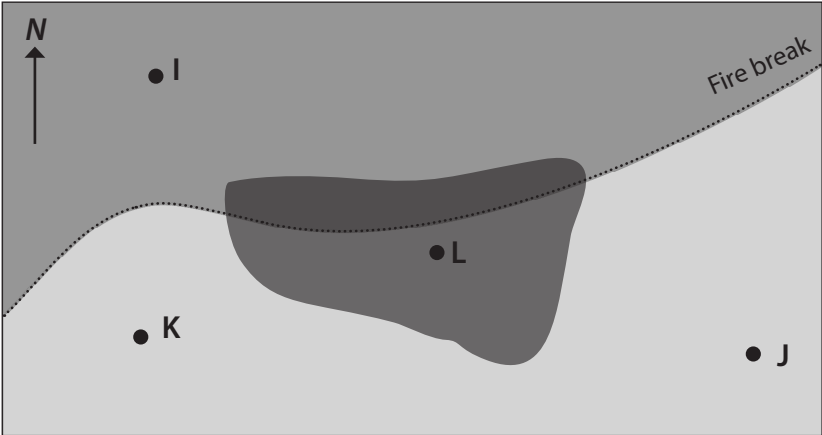
Common, ground-dwelling bird

a Some differences between the birds have been recorded in the table below. Complete the table by identifying and describing two more features that help to tell the two birds apart.

FEATURE	DESCRIPTION – RARE, GROUND-DWELLING BIRD	DESCRIPTION – COMMON, GROUND-DWELLING BIRD
Beak	Wide at the base and dark in colour	Wide at the base and light in colour
Head		
Legs		

b What is the actual height of the rare, ground-dwelling bird?

1.2 The map below shows a section of the national park that was investigated by scientists in January 2005. The graph below shows how the ‘percentage of ground covered’ changes after fire for the three types of plants.



KEY

.....

Fire break

■

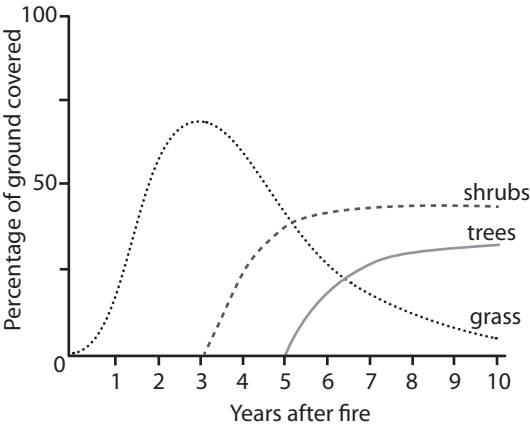
Area burnt, Jan. 2001

■

Area burnt, Jan. 1998

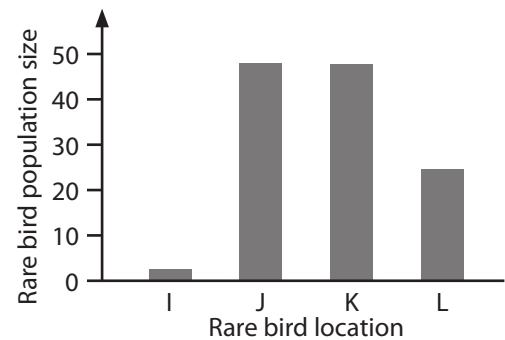
■

Area burnt, Jan. 2004



- a Describe the vegetation that the scientists would see at location L in January 2005.
- b In which year would you expect to see at least 25% trees in location L?

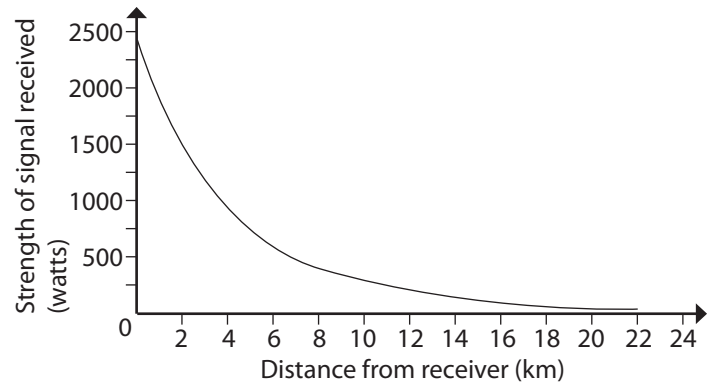
1.3 In January 2005, the scientists estimated the size of the rare bird population at four different locations (I, J, K, L) in the park. The population sizes are shown in the graph at the right. Assuming no further fires occur in the park, predict the size of the population of rare birds at location I in January 2008. Explain your answer.



The scientists investigated the movements of one of the rare birds through the park by using a radio transmitting device attached to its wing.

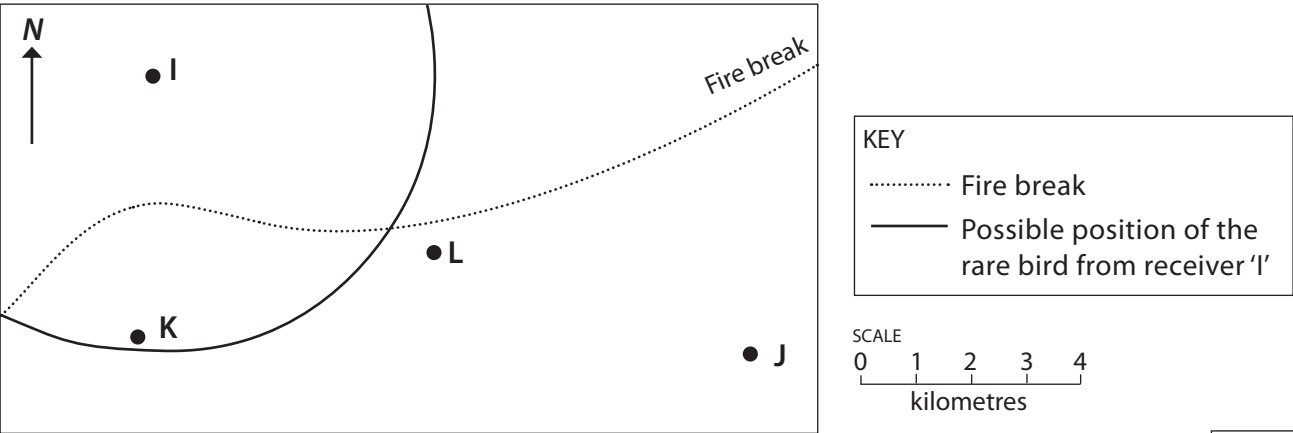
This transmitting device produced a radio signal that could be picked up by receivers positioned throughout the park. The graph below shows the relationship between the distance from the receiver and the strength of the signal. The strength of the radio signal was used to work out the distance the bird was from a particular receiver. Signal receivers were set up at three locations (I, J, K). The results were recorded by the scientists in the following table.

Receiver location	Strength of the signal (Watts)	Distance from the receiver (km)
I	750	5
J	350	
K	980	



1.4 Using the results from location I, the scientists constructed a curved line on the following map. The scientists know that the rare bird is located somewhere close to this line.

- a There are some missing results in the table. Use the graph to help you complete the table.



- b Using these results and a ruler, construct two more curved lines on the map above to show the possible position of the rare bird from signal receivers at J and K.
- c Shade the area on the map that shows the most likely position of the rare bird.
- 1.5** An animal with a transmitter is travelling away from the receiver at J. At about what distance from this receiver would the strength of the signal received at J become zero?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

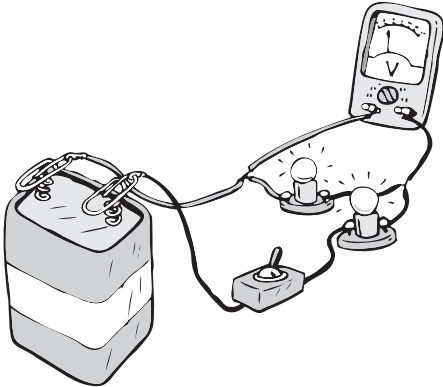
Diagrams vs Sketches

In science, communicating ideas and information may involve visually displaying the properties or results of an investigation through diagrams and sketches. Diagrams represent relationships between things. Sketches represent the structures and form of objects.

Diagrams

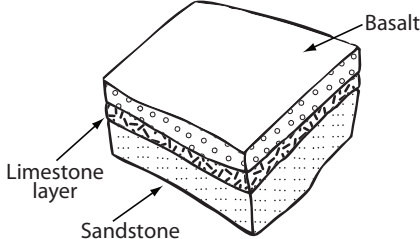
Activity 1

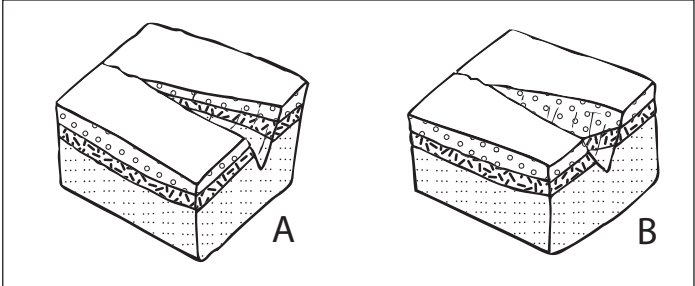
- 1.1 Electric circuits are often drawn as circuit diagrams. Use the key shown to redraw this electric circuit as a circuit diagram.
- 1.2 The diagram below shows the layers of rocks that make up a plateau near the coast. Which diagram below right, A or B, best represents the limestone layer if a valley is eroded away from this plateau? Give a reason for your answer.



KEY

- Globe
- Switch
- Voltmeter
- Wire
- Power supply





Maps and charts

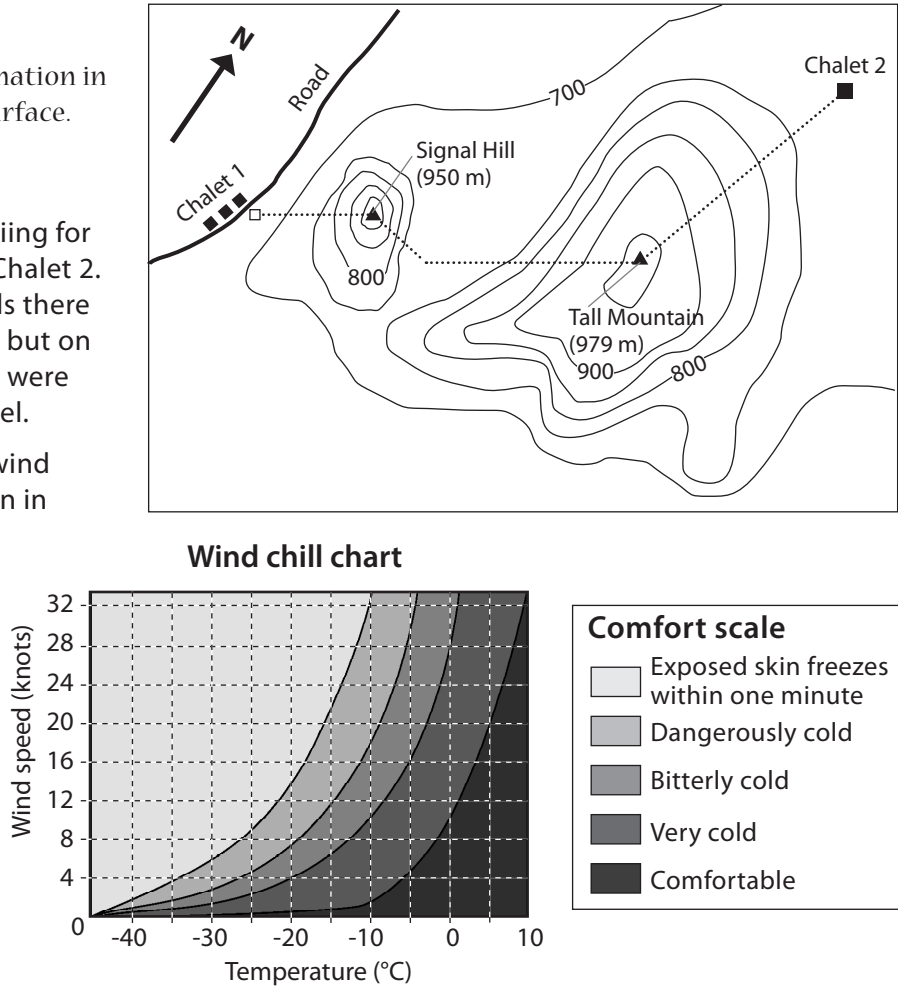
A map or chart presents accepted information in a methodical way and on a horizontal surface.

Activity 2

- A group of friends went cross-country skiing for the day. They travelled from Chalet 1 to Chalet 2. In the flat areas and on the top of the hills there was no vegetation above the snow level, but on the north- and south-facing slopes there were low trees and shrubs above the snow level.
- A wind chill chart shows the effect that wind speed has on the comfort of people when in low temperature areas.

2.1 In the wind chill chart:

- a What is recorded on the x-axis?
What units are used?
- b What is recorded on the y-axis?
What units are used?



2.2 One skier noticed that on the flat land, the temperature was 0°C . Halfway up Tall Mountain it was very windy, bitterly cold and the temperature was -15°C . At the top it was still the same temperature but felt much colder than halfway up.

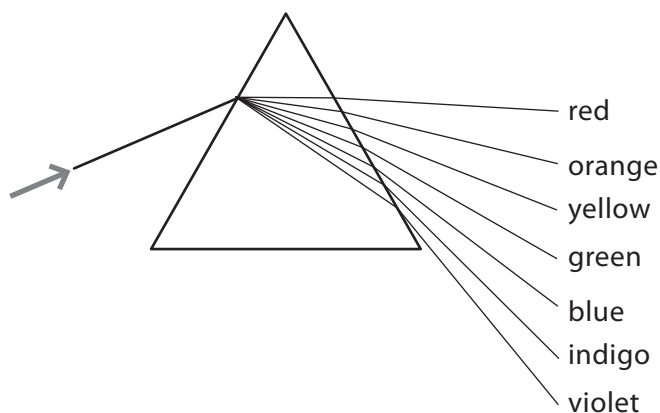
- Use the Wind Chill Chart to estimate the wind speed range halfway up the southern slope of Tall Mountain.
- Explain why it felt colder at the top of Tall Mountain than it did halfway up.
- Predict what the wind chill will be at Chalet 2 if the wind is comfortable and the temperature is 12°C .
- If the temperature at the top of Tall Mountain fell to -20°C and the wind speed rose to 45 knots, what would be the comfort rating?

Using and interpreting lines on diagrams and sketches

Sometimes lines and arrows on diagrams are not supposed to be interpreted literally but rather as a 'commentary' or as explanatory 'notes' for the diagram.

Activity 3

White light bends when it passes through glass. White light is made up of many colours. These colours all travel at the same speed through air. However, when they pass through glass they slow down by different amounts and are refracted through different angles. The smaller the change in speed, the smaller the angle of refraction.

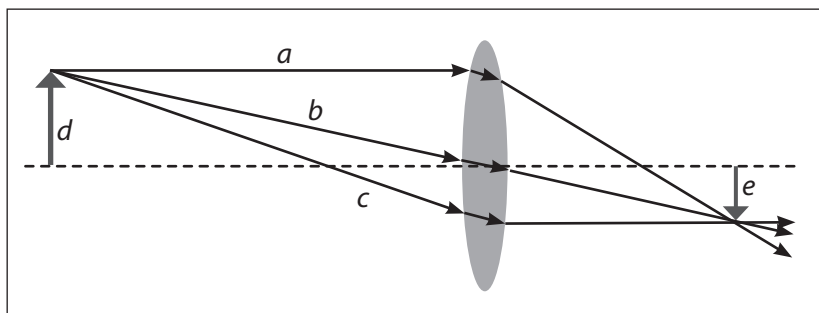


3.1 In the diagram of the glass prism:

- what do the lines represent?
- what does the arrow represent?
- according to the lines, which colour slows down the least?

3.2 In the diagram of light passing through a glass convex lens:

- what do arrows *a*, *b* and *c* represent?
- what does arrow *d* represent?
- what does arrow *e* represent?
- why is arrow *e* pointing down?



Sketches to scale

The scale of a drawing relates the size of the drawing to the size of the real object it represents. The two most common ways of writing a scale are:

- ♦ **unit scale** such as 1 cm : 1 m – that is, a 1 cm length on the drawing represents 1 m in reality
- ♦ **size scale** such as 1 : 100 – that is, the real object is 100 times larger than the drawing.

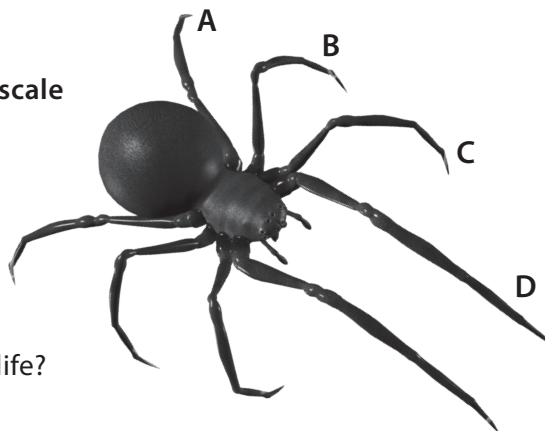
Activity 4

4.1 On the scale drawing of the spider, if the **unit scale** is 1 cm : 2 mm:

- how long is the spider's body?
- which leg is 5 mm long in real life – A, B, C or D?

4.2 If the **size scale** is 1 : 10:

- how much bigger would the spider be in real life?
- how long would leg D be in real life?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Why use graphs?

Graphing is an essential communication tool. Graphs are an information-dense system compared with verbal or written communication. They fit a lot of information into a small space, so that data can be visually compared and relationships identified.

Comparing two variables – scattergram

If you want to compare two variables, which might not have a dependent relationship, you can draw a scattergram. In a scattergram, each point on the graph represents a pair of two variables – one coordinate from the variable on the x-axis, and the other coordinate from the variable on the y-axis. The points can be joined by a curve or by a line of best fit. The curve or line passes through the mean of each set of data, so that there are approximately the same number of points above and below it. This emphasises trends in the data and shows an approximate relationship between the two variables.

Activity 1

1.1 The data in the table below represents the relationship between the height and the mass of a group of netball players.

HEIGHT (cm)	151	152	152	154	152	154	154	155	156	157
MASS (kg)	50	50	51	52	53	54	56	56	57	57

- a Draw a scattergram to show the relationship between the two variables.
- b What is the mean of each set of data?
- c Draw the line of best fit on the graph.

Comparing multiple variables

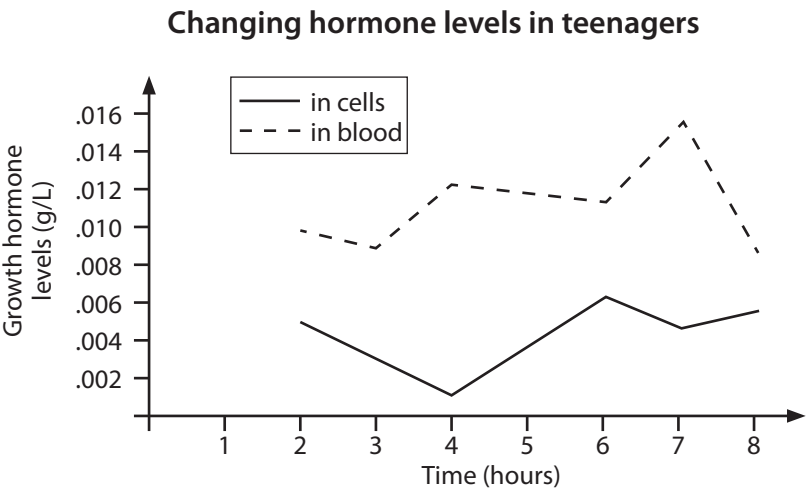
Graphed information becomes more complex as the number of variables presented in the graph increases. However, plotting two or more sets of data on the same graph is a powerful tool in science. It makes it easy to see the differences between the graphs as well as how one graph relates to the other.

Additional variables are often represented as multiple lines on a graph. If there is more than one line on a graph or if symbols are used, then a key must be provided so that each line or symbol is easily identified.

Activity 2

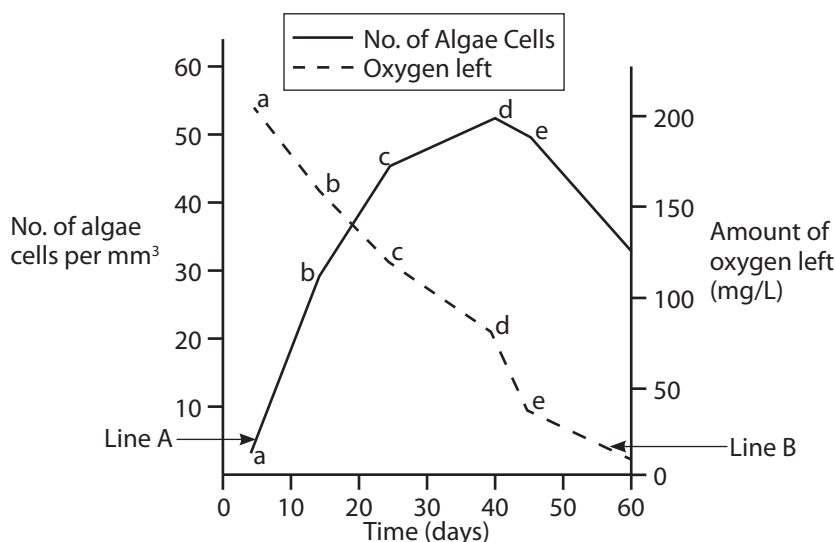
2.1 When more than one line is used on a graph, sometimes you can use the same scale to draw the two graphs. The example below uses the same scale on the y-axis to compare growth hormone levels in teenagers.

- a Is there a key provided?
- b What is the difference between the hormone levels in cells compared to the hormone levels in the blood at:
 - ▲ 2 hours?
 - ▲ 4 hours?
 - ▲ 6 hours?
- c What trend can you see between the hormone levels in cells and the hormone levels in blood?



2.2 If the two variables have different units, then different scales must be used to plot the data – one scale on the left side and another scale on the right side. The example at the right uses two different scales to compare the effect of algae on oxygen levels in a pond.

Effect of algae on oxygen levels in a pond



a At what oxygen levels did algae growth start to slow?

b How many days does it take for algae to start dying?

c What is the general relationship between algae numbers and oxygen levels?

d Once the algae started to die, what happened to the oxygen levels?

Tools for decoding a graph

The more complex the graph, the more concepts can be related simultaneously. We convert the visual image of graphs into meaningful information by looking for relationships and trends between the independent and dependent variables. There are several tools that help us to do this analysis.

Activity 3

3.1 Working out the position and value for a point on a line between two known points is one way of predicting the unknown value fairly accurately.

a How many algae cells were in the pond on day 55?

b What was the oxygen level on day 5?

3.2 A line moving away from both axes shows a positive increase in both variables. These variables are directly proportional. When a line or curve moves downwards towards the bottom right corner, it indicates that change is decreasing.

a Which line, A or B, demonstrates that the rate of change is always decreasing?

b Where on line A is a positive directly proportional relationship shown between the independent and dependent variables?

3.3 The slope of a line is called its gradient. The gradient shows how fast the data changes and whether there is a direct or inverse relationship between the two variables. Consider the lines in the algae graph above. A straight line indicates a constant change. The steeper the slope, the greater the rate of change.

a Where on line A is the rate of algae reproduction the greatest?

b Where on line B is the rate of the oxygen levels falling sharply?

c Describe what is happening between points c and e on line A?

d Describe the overall trend of line A.

3.4 Extrapolation is a process of predicting a point beyond the available data on a graph. Of course, these predictions are not as reliable because you do not know if the slope of the line will change. To make a prediction, we must assume that the slope will remain the same as in the last segment.

a Assuming oxygen levels were kept constant, predict how many algae cells would be present in the pond after 30 days.

b If the slope of section c to d on line B stayed the same, what would be the level of oxygen in the pond on day 60?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Data analysis, evaluation and conclusion

Science puts observations to the test in experiments that refine and evolve new theories.

Scientists are constantly comparing things in their search for similarities and differences. They look for the patterns and codes that may uncover new relationships and they propose simple sets of rules that relate to a wide range of observations.

Much of science can be trial and error

Scientists devise theories then test them. If the test doesn't work, they adapt or change the set-up. Then they try the test again – and so on.

As an investigation is carried out, scientists note where improvements can be made. They always record any changes to the original method or the materials required. This way, other people are aware of and can use the modified experimental procedure.

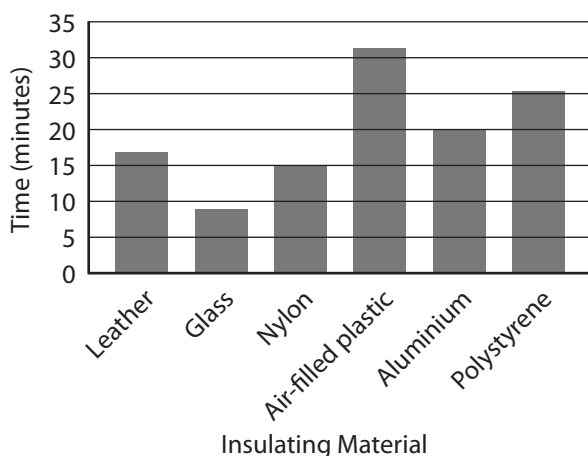
Scientists look for sources of error in the method as well as in the way data is recorded, and note any sources of error or ideas for improvement so they can be applied to further experiments.

Activity 1

Cans of cold soft drink were each placed into similar containers made of five different materials: leather, glass, nylon, air-filled plastic, aluminium and polystyrene.

The time taken for the temperature of the soft drink to rise by 5°C was recorded for each can. This data is presented in the column graph below.

Time taken for a cold can of soft drink in a container of insulating material to heat up by 5°C



1.1 Refer to the column graph to answer these questions.

- How many insulating materials were tested?
- Which material allowed the soft drink to heat up the fastest?
- Which material kept the soft drink cold for the longest time?
How long did it take for this can to heat up by 5°C ?
- How much longer did the can in the aluminum container take to heat up than the can in the leather container?

1.2 Consider the evidence and rank the materials in order from the worst insulator to best insulator.

1.3 Interpret the evidence by extracting a supportable conclusion from the data. Which is the best insulator?

- 1.4 Inferences** are suggestions based on observation. Sometimes it is necessary to infer beyond the data collected. These inferences are based on logical deduction and a sense of probability that the inference is or is not true. Make an inference about the data – why do you think your chosen material is the best insulator?
- 1.5 Scientific knowledge** is used to explain an inference or a valid conclusion. Answer **a** and **b** to explain your conclusion in question 1.3.
- a** What scientific information allows you to make this inference?
- b** Suggest why polystyrene is also a good insulator.
- 1.6 Recommendations** for changes in experimental design can be made at this stage. These changes could be to improve the method or the way that data was collected. Would you change the experimental design of this experiment? If so, how?
- 1.7 Sources of error** are things that could affect the experimental outcome and should be recognised. They include poor use of controls, uncontrolled variables and any errors in reading meters and dials – which is commonly called human error. List three places in this investigation where error could be minimised better.
- 1.8** Scientists make **suggestions** about other types of experiments that might relate to what they have done, and might be useful to other researchers. Write two ideas for related experiments that could expand this line of research.

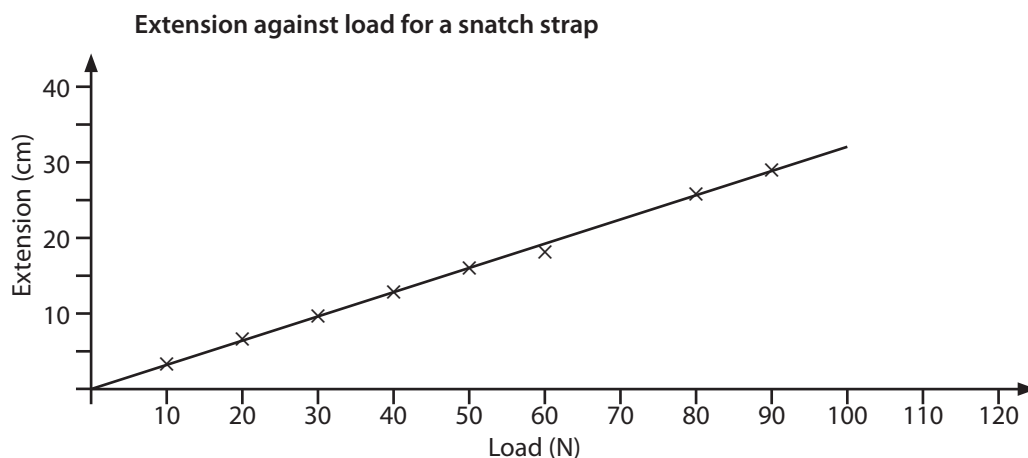
Deciding what you think your results show

A conclusion is a summary of what you found out, and sometimes what you didn't find out.

Well processed results show patterns and relationships which make it easier to reach a conclusion. Compare the conclusion with the hypothesis – do the data agree or disagree with your original guess? Hypothesis are never right or wrong; they are just supported or not supported. In both cases, you learn something.

Activity 2

- 2.1** Study the graph, which shows the relationship between extension and load for the snatch strap on a 4WD vehicle. A snatch strap is a type of tow rope.



The hypothesis being tested is: *Snatch strap extension is directly proportional to the load.*

- a** What conclusion could be drawn from this data?
- b** What is the rate of extension of the snatch strap?
- c** Does the conclusion support the hypothesis? Explain your answer.

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



What scientists learn from each other

Scientists discuss their findings with others so that the original ideas can be built upon, and new ideas, possibilities, developments and potential problems can be considered. Through this need to exchange knowledge, scientists have created impressive, world-wide networks for communication.

Science reports – organising the information

A science report is generally divided into different sections in order to clarify the information and to make it easier to read and easier to compare with other reports. Reports are read by a variety of audiences, so the use of a recognised format assists the reader in understanding the investigation, the data and the implications.

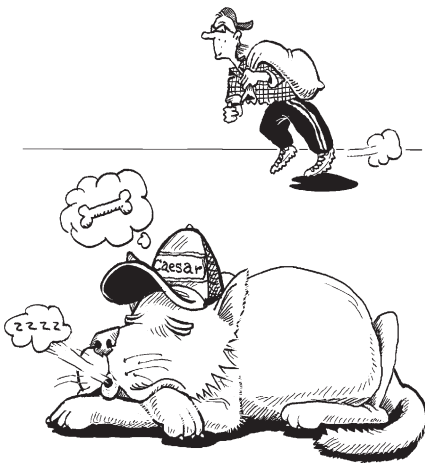
Activity 1

1.1 Consider the terms in the table below and briefly describe the differences between each pair.

WHAT IS THE DIFFERENCE BETWEEN ...		
a	a hypothesis and a conclusion?	
b	a conclusion and an inference?	
c	the results and the discussion?	
d	the introduction and the literature review?	
e	the equipment section and the method section?	

1.2 Identify each statement below as an *observation* (O), a *hypothesis* (H) or a *conclusion* (C).

- a The dog in the house down the road is barking._____
- b There are no cars in the driveway._____
- c The owners must be away._____
- d There could be someone walking by._____
- e I heard the muffled sound of conversation._____
- f The dog is still upset._____
- g Someone is assessing the house._____
- h They may break into the house tonight._____



A valid conclusion

Being able to draw a valid conclusion means that you have carefully considered questions, such as:

- What were the results?
- What do the results mean?
- Are the results useful?
- What did we learn?

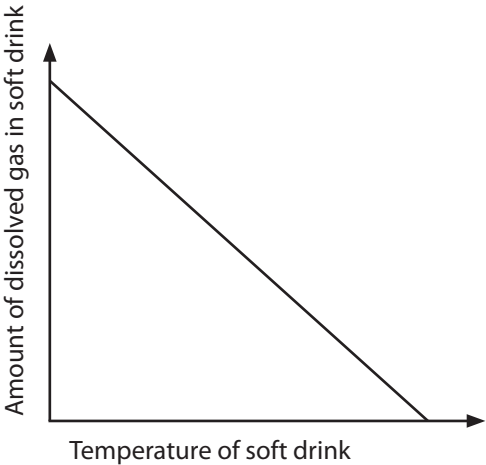
At this point, there are usually even more questions to answer – such as *Where does this information lead us?*
What do we need to investigate further?

Activity 2

An important skill in science is the ability to draw reasonable conclusions and make objective judgments.

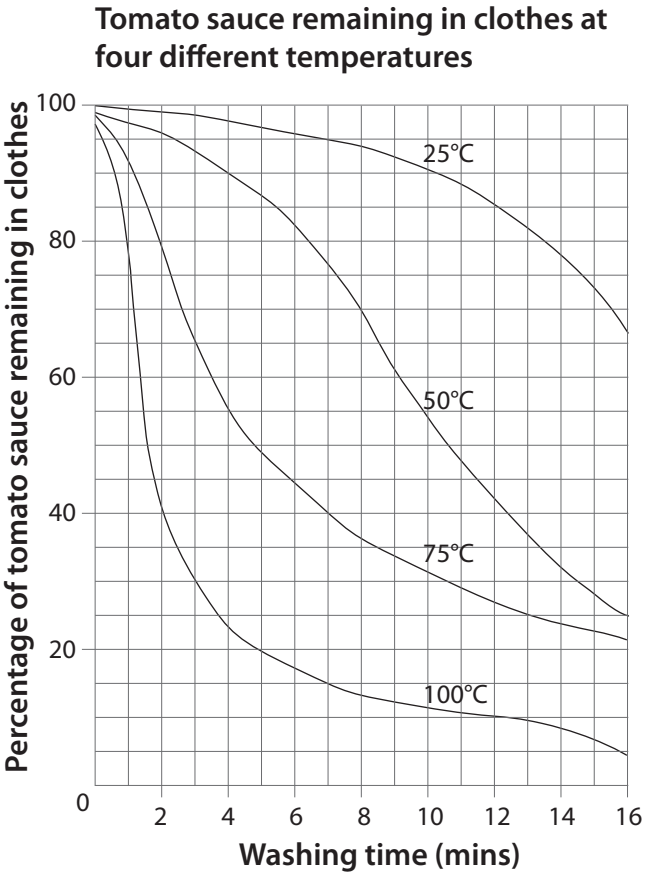
2.1 The graph at the right shows the amount of dissolved gas in a can of soft drink as the temperature rises. Tick the correct conclusion.

- a The amount of gas does not change when the temperature changes.
- b The amount of gas increases when the temperature increases.
- c The amount of gas decreases when the temperature decreases.



2.2 The graph at the right shows that after 10 minutes of washing in water at 75°, 30% of tomato sauce stains remain. Consider the other information shown in the graph. Which conclusion is the most reasonable for these results?

- a Temperature does not affect how quickly tomato sauce stains are removed.
- b Tomato sauce stains are washed out faster at high temperatures.
- c Tomato sauce stains can be removed after 30 minutes of washing time.



2.3 Trucks often have large signs above the cabin to advertise the name of the company. Study the data below and then write a reasonable conclusion.

Height of sign above the cabin (m)	1	2	3	4
Time (min) to travel 1 km	0.4	1.4	2.4	3.1

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Safety in a laboratory

A laboratory environment has particular hazards – for example, electrical tools, glassware and chemicals can be quite dangerous if used incorrectly. Some experiments that scientists carry out in a laboratory are potentially lethal. It is important that experimenters handle equipment, organisms and chemicals properly, carefully and safely.

It's all about common sense!

Safety in a laboratory is largely common sense. Do you use your common sense when you are in a science laboratory?

Activity 1

1.1 Do you break any of the rules when doing an experiment? Answer honestly! Are you guilty of:

- a** working on a dirty or cluttered bench?
- b** leaving bags lying around on the floor?
- c** neglecting to wipe up spills immediately?
- d** not replacing stoppers in reagent bottles?
- e** not cleaning things and putting them away when you have finished with them?
- f** wandering around the laboratory, especially when carrying chemicals or glassware?

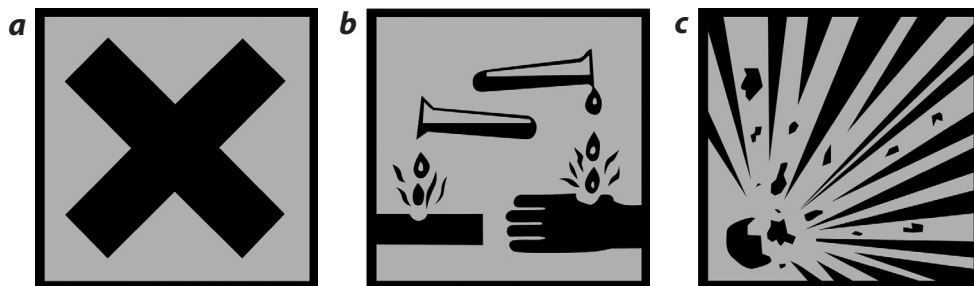
1.2 List the potential dangerous consequences of each behaviour described in question 1.1.

Signage

To help to reduce risks in laboratories, standard signage has been developed. There are important hazard warning labels that are commonly found in school science laboratories.

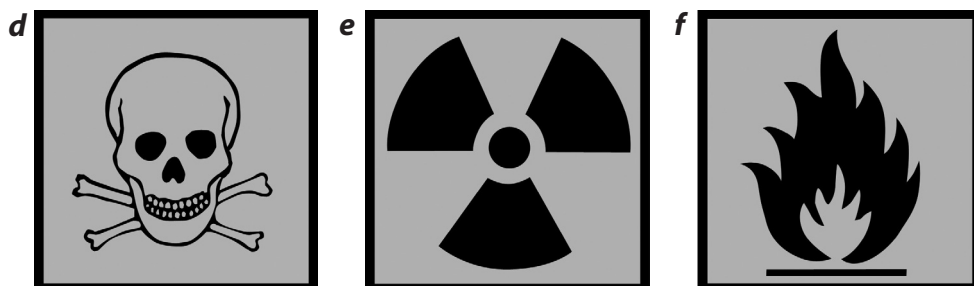
Activity 2

2.1 Write the correct word above each of the hazard warning signs: *corrosive, explosive, flammable, harmful, toxic, radioactive*.



2.2 Circle the signs that are displayed in your laboratory, or for which there are similar ones with the same purpose displayed.

2.3 Describe any other warning signs in your laboratory and briefly explain the reason for each one.



Chemicals

A Material Safety Data Sheet (MSDS) is a fact sheet which provides information that allows the safe handling of that particular material. MSDS are provided by the manufacturer.

Activity 3

3.1 A MSDS provided the data shown at the right about a commonly used substance in a school laboratory.

- a What do these words mean?
▲ flammable ▲ toxic ▲ corrosive
- b Will this substance burn?
- c What protective clothing and equipment should you use if you are handling this chemical?

MSDS DATA

Flammable limits – Not flammable, if heated to high temperatures it will emit toxic fumes
Extinguishing media – Does not burn nor will it support combustion
Swallowing – Ingestion can cause diarrhoea, vomiting and stomach pain
Skin/inhalation – Some people are sensitive/irritation to upper respiratory tract
Eyes – Severe eye irritant, causes substantial, temporary eye damage, may be corrosive to eyes
Clothing – Causes permanent stains

3.2 Draw lines to match the safety phrases from chemical containers with the correct meanings.

- a Explosive
- b Harmful if swallowed
- c Do not inhale vapour.
- d Avoid contact with eyes.
- e Toxic by contact with skin
- Do not get any of it in your eyes.
- Poisonous if absorbed through the skin
- Do not breathe in gases produced.
- May explode in the right conditions.
- Don't eat it.

Fire in a laboratory

If there is a fire in your laboratory, you cannot just pick up the closest fire extinguisher to put it out – you could make things much worse. You should not even use a fire extinguisher unless there is no other option. There are many different fire extinguishers and they are colour coded. The main types of fires and the extinguishers to use are listed in the table below. They all operate by smothering the flame or denying the fire oxygen.

KEY: ** most suited. * may be suited. X not suitable.

TYPE OF FIRE EXTINGUISHER	WATER	FOAM	CARBON DIOXIDE	DRY CHEMICAL	VAPORISING LIQUID
Type of fire	Red	Blue	Red/ black band	Red/white band	Red/yellow band
A. General – wood, paper, fabrics	**	*	*	*	*
B. Flammable liquids – paint, chemicals	X	**	*	*	* (not fat or oil)
C. Flammable gases – hydrogen, ethylene	X	X	X	**	*
D. Combustible metal – magnesium, sodium	X	X	X (sand in an emergency)	X (sand in an emergency)	X
E. Electrical – sparks from current flow	X	X	*	*	**
F. Fat and oil	X	X	*	*	X

Activity 4

4.1 Which extinguisher would you use on the following fires?

- a Waste-paper basket
- b Burning sodium
- c Oil
- d Electrical fire

4.2 What type of fire extinguishers can you find in your laboratory?

Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.



Comparing models with the real world

There are many concepts that scientists cannot easily discover from first-hand experience. A model is a tool for learning about the things that they are meant to resemble. A model may take many forms, including diagrams, mathematical formulae, physical constructions or even a set of logical statements.

We can use models to represent phenomena that are too large, too small, too difficult or impossible to see, or present a hazard. Models help us to grasp incomprehensible concepts such as geological time and the enormity of space, as well as infinitesimally small ones such as atoms.

Activity 1

1.1 The table below lists five real world processes or situations for which models are often used in science investigations. Draw lines to match each process with the best example. Then write a second example for each one.

REAL WORLD PROCESS THAT SUITS A MODEL	EXAMPLE
a The process happens too slowly.	• What will happen if the angle of the Earth’s axis is changed? •
b The process happens too quickly.	• Will the new drug cure the disease? •
c The process is too small scale to observe directly.	• What would the atmosphere be like if its molecules were to act like tiny, speeding marshmallows instead of tiny, speeding hard balls? •
d The situation is too vast to be changed deliberately.	• What happens to a dummy in a head-on car crash? •
e The process is potentially dangerous.	• Will the genetic defect ‘die out’ with enough generations? •

Adopting a questioning attitude

When considering the conclusions reached by modelling a phenomena, data validity and reliability must be taken into account. In other words, the behaviour of a model cannot ever be expected to represent a full-scale situation with complete accuracy. A model might only have limited usefulness. The limits of modelling must always be respected.

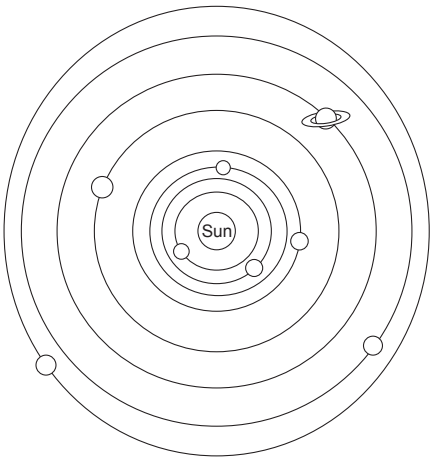
Activity 2

2.1 For each limitation in the box below, write the letter beside the best example in the following table.

LIMITATIONS		
a Changes in scale or the presence of differences that are not taken into account in the model	d Powerful computers play an enormous role in modelling forecasts; however, they do not include the critical 'human factor.'	g Type of wear that is unlikely to occur over the life of the shoe
b 'Slow-down' models may not accurately demonstrate the overall movement.	e Water flow is different in the ocean.	h Larger doses may have different effects (e.g. kill instead of cure) not just quicker effects.
c A model may be too complex for practical use. Accuracy may be limited by the large number of interacting variables that need to be dealt with simultaneously.	f 'Short run' models may attempt to compress long-term effects by increasing the rates at which events occur.	i Electric currents are not wet and if a circuit gets a hole in it, the electrons would not leak out like water from a hose.

EXAMPLES OF MODELS	LIMITATIONS
Performance of a model boat in a tank	
Hastening the investigation by using larger doses of drugs	
The destructive testing of products – using machines to simulate the wear on shoes	
Using a model of water flowing through a hose to describe electrons flowing through a circuit	
Computer-produced predictions of such phenomena as global weather, El Nino, volcanoes, tidal waves, etc.	
Rats may be sensitive to a drug that people are not.	
Complex models of such things as global population growth, weather and global food supply	
Recording the very rapid motion of birds and dancers	
Giving massive doses of chemicals to laboratory animals to try to get in a short time, the effect that smaller doses would produce over a long time	

- 2.2 Think of a model used or discussed in your science classroom.
- a What phenomena did the model represent?
 - b What were the limitations of the model?
 - c Did the use of the model increase your understanding of the phenomena? Explain your answer.
- 2.3 Study the model of the universe shown.
- a What features of the universe can be explained usefully with this model?
 - b What are the limitations of this model?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

The world according to a scientist

Scientists share certain basic beliefs and attitudes about what they do and how they view their work. Science presumes that events in the universe occur in patterns that can be trusted and are understandable. Scientists also acknowledge that science cannot produce answers for everything, but that the knowledge it does produce is durable.

The rules of the game

Science assumes that the universe is a vast system in which the basic rules are the same everywhere. This means that if we study one part of the universe then the information gained can be applied to other parts of the universe. They believe that these 'rules' can be discovered by careful systematic study and also that the rules are ever-changing because science is a human discipline and there are no facts. The most important rule in science is to avoid jumping to hasty conclusions but still recognise when a theory has outlived its usefulness and should be replaced or updated.

PROOF

A proof is a scientific demonstration that something must always be true. Most proofs involve mathematics. Unlike a theory or a scientific law, a proof is not liable to be changed or replaced as more observations are made.

THEORY

A theory is a generally accepted scientific explanation that seems to fit all the observations made about a subject, and seems to show how or why something happens.

LAW

A law of science is a statement or explanation that always appears to be true. A scientific law or principle is a theory that has been thoroughly tested, and shown to be true. It explains things that have happened and also makes predictions about events that have not yet taken place.

Activity 1

- 1.1 What do we mean when we say that scientific knowledge is 'durable'?
- 1.2 Do scientists believe that they can find a solution to every problem?
- 1.3 Is there such a thing as 'scientific fact'? Explain your answer.
- 1.4 What is the advantage of believing that the basic rules of the universe are the same everywhere?
- 1.5 Are the following notions proofs, theories or laws? Fill in the correct word.
 - a Hooke's _____ states that current is proportional to voltage.
 - b Maths was used for a _____ about how many cells would be produced when a single cell divides a set number of times.
 - c Darwin's _____ of evolution seems to explain why living things slowly change as time goes by.
 - d Sir Isaac Newton's _____ of motion and gravitation
 - e Edwin Hubble's Big Bang _____ describes the birth of the universe.

- 1.6** Use these words to complete the information about the changing nature of science: *observation, theory, theories, truth, knowledge, new, true, challenge, law*.

Science is a process for producing _____. The process depends on careful _____, then inventing a _____ to explain the observations. Change in knowledge is inevitable because _____ observations may _____ current theories. Testing, improving and discarding _____, whether new or old, goes on all the time. Scientists assume that there is no way to secure absolute _____, just to make more accurate approximations. A _____ is a theory that has been thoroughly tested and appears to be always _____.

Seeing force and motion

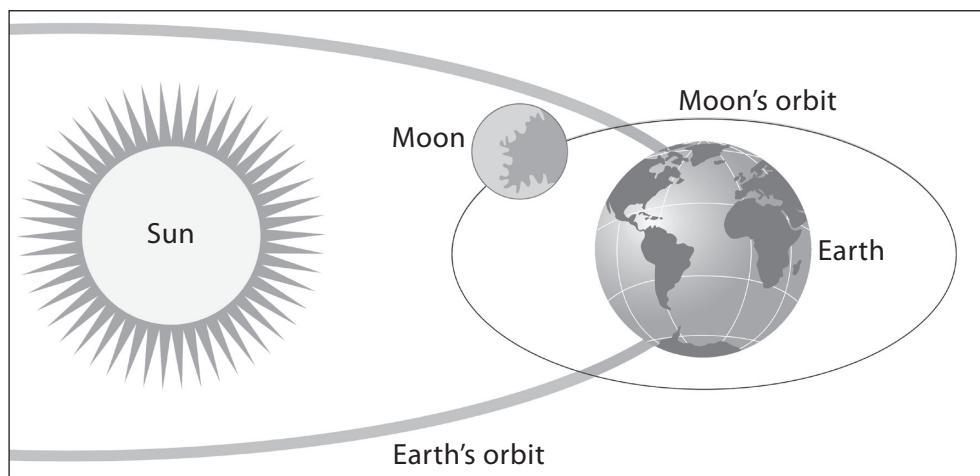
Isaac Newton proposed one set of simple, mathematical laws for motion on Earth and in the sky. These laws showed that the universe ran automatically, according to the action of forces between its parts.

Newton created a unified view of force and motion in which motion everywhere is explained by the same few rules. He analysed gravity and motion and showed that planetary orbits had to be the ellipses that Johannes Kepler proposed at the time Galileo was studying the solar system. Newton's laws are based on the observation that the force of gravity between any two objects in the universe depends only on their masses and the distance between them. This mathematical model made it possible to account for such things as the tides, the orbits of planets and moons, the motion of falling objects and the Earth's equatorial bulge.

For several centuries Newton's science was accepted because it explained so many different phenomena and could be used to predict physical events such as Halley's comet.

However, in the 18th century, philosophers outside science were concerned that if all motion runs according to mathematics, then the human concept of 'free will' might only be an illusion.

Despite being overtaken in the 20th century by Einstein's relativity theory, Newton's work is still valid today for its sheer 'explaining' power.



Activity 2

- 2.1** Use the information above to answer the following questions.
- Which scientist predicted that the planetary orbits were elliptical before Newton proved it mathematically?
 - What is the basis for Newton's model of motion?
 - Provide two examples of where Newton's laws have been used to explain other phenomena.
 - Powerful ideas in science tend to survive and grow more precise. Which new theory included and extended Newton's laws rather than rejected them?
 - Although Newton's ideas were well accepted by the scientific community, why were 'social thinkers' troubled in the 18th century?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Know what science is about and how to use it!

Scientific literacy means being able to understand the language of science and use basic scientific skills to make intelligent decisions. This enables us to keep track of developments throughout our lives. Science is not something that is done to us, or for us by experts – there is an opportunity for us as members of society to get involved with it.

Scientific literacy

Being scientifically literate means you can contribute to the way you want the world to develop because you have a basic understanding of science and how it works. The knowledge and understanding that exists in scientific journals is only potential human knowledge. There are many factors involved in transforming these ideas into an everyday reality.

Activity 1

- 1.1 Use your basic scientific skills and knowledge to make intelligent decisions about the following current issues. For each one, state your personal viewpoint and list three scientific considerations that helped you to decide.
- a Should nuclear power stations be built in Australia?
 - b Should desalination plants be built to provide water to large cities?
 - c Should stem cell research be allowed?
 - d Should clean coal technologies be used for power, or would it be better to invest in developing ‘green’ power technologies?
 - e Should we clone animals? Humans?

Valid reporting

Science is exciting, and media reporting keeps people informed of developments. However, sometimes what is presented as ‘science’ in the media is inaccurate and not properly qualified. Poor reporting damages the credibility of science and opens the way for superstitions and unfounded beliefs.

Of any announcement about any topic, we should ask ‘Is that likely to be so?’ With science announcements, a scientifically literate person will be able to ask ‘Is it really so?’ Speaking with confidence about science requires the understanding of some of the basic science concepts.

Activity 2

- 2.1 For each headline in the table, decide whether it is likely to be so, and if it really is so. Briefly explain the reasons for your answers.

HEADLINE	IS THAT LIKELY TO BE SO?	IS IT REALLY SO?
Space tourism		
From the sewer to your glass!		
First cloned human born		

Fear – generating sales

Science involves mental discipline. If people cannot think in a scientific way, they may blindly accept all that they are told, and believe a confusing mix of science, superstition and popular judgment. Consider how society reacts to media headlines that are geared towards frightening people, in order to get their attention. Where is the line between responsibly warning people of potential problems, and provoking unnecessary fear and paranoia?

Activity 3

3.1 Mad cow disease and Frankenstein foods (genetically engineered) have sparked a worldwide consumer backlash against genetic modification.

What is your opinion of the media’s coverage of these issues? Do you think ‘a worldwide consumer backlash against genetic modification’ is a reasonable reaction? Explain why.

3.2 Biological weapons from genetic research? Will genetic engineering give rise to genetically engineered bio-weapons that target specific ethnic groups?

Is the fear expressed in this headline well-founded from a scientific, social, economic or political point of view? Explain your opinion.

3.3 Draw lines to link each headline with the way some people reacted to it.

HEADLINE	HOW DID SOME PEOPLE REACT?
a Aluminum linked to Alzheimer’s disease	Adopted paranoid behaviour about sunlight
b Beaches lost and wetlands destroyed as sea levels rise	Started stockpiling gold in case of a global economic collapse
c Margarine causes breast cancer	Stopped using aluminium cookware
d Global warming has become critical to our economy	Live in fear of imminent environmental disaster
e UV light causes skin cancer	Stopped eating margarine

3.4 How could people who reacted as listed in the previous question avoid living in fear and stress? Read the suggestions below and tick the ones that you, as a scientifically aware individual, think would help.

- a Start asking questions about the claims.
- b Get on the Internet and read scientific reports on the issues.
- c Find out who did the research and in what scientific journal it was originally published.
- d Pretend it’s not happening and get on with life.
- e Find out who funded the research – did they have a financial interest in the results?
- f Engage in conversation with people who are experts in the field.
- g Don’t read any more newspapers and sell the television.

3.5 Now extend your answers from the previous questions. Choose one headline listed in question 3.3. Following the suggestions from question 3.4 that you think would help, compile a scientifically aware report about the issue that will present balanced information to someone who has no scientific understanding of it.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Real decisions from real science

Experimental science is motivated by curiosity about the world. People want to gain knowledge about how the world works, and find ways to change, use and improve it. However, while scientific discoveries have the potential to provide great benefits to humans and the world, some discoveries also have the potential to cause great destruction, both environmental and social.

Activity 1

1.1 Think about the scientific discoveries people have made over the last 100 years.

- a** Write a list of the ten scientific discoveries you think have been most beneficial to people and the world.
- b** Now write a list of the ten scientific discoveries you think have caused the most damage to people and the world.
- c** Consider the lists you wrote. Write a paragraph explaining your reasons for choosing the discoveries on each list.

The conundrum of human transplants

Current ethics in science hold that research involving human subjects may be conducted only with the informed consent of the subjects, even if this constraint limits the research or influences the results.

Activity 2

FETAL CELL TRANSPLANTS

Some day soon, the living cells of unborn human babies may keep other people alive, healthy and youthful. Fetal cells at the proper stage of development can reach maturity and function normally if transplanted into the organs of humans. People with cancer, degenerative diseases such as insulin-dependant diabetes and injuries which confine them to wheelchairs could receive a fetal cell transplant operation.

Today, the use of fetal tissue for any purpose stimulates strong emotional responses in relation to moral issues, religious beliefs and the status of human embryos.

2.1 Under what circumstances would you accept the cells of an unborn fetus?

2.2 What limits, if any, should be imposed on fetal tissue transplant research and development?

2.3 For each statement below, decide whether you agree or disagree, and explain the reasons for your choice.

- a** The replacement of degenerated organs and tissues will extend the human lifespan to a point that is beyond the best interests of the species.
- b** Injecting fetal cells into the facial skin of people who want to look younger should never be allowed.
- c** Injecting fetal cells into the facial skin of people who have experienced injury that has caused severe disfigurement and scarring should never be allowed.
- d** People should be allowed to benefit financially from selling fetal tissue and other types of human tissue as this encourages more research and development in this area of science.

2.4 A young boy has incurable cancer. His parents want to select a specific embryo so that they can have another baby with the same genetic make-up. This way, the baby's blood can be used for blood transfusions to cure its older brother. Should this be permitted?

2.5 The twin sister of a woman suffering from renal failure has offered to donate one of her kidneys. Should the kidney be transplanted into the woman to save her life?

Appropriate use of science

The ethics of science is also concerned with the possible harmful effects of applying the results of research. The long term effects of science may be unpredictable, but an idea of the possible applications scientific work can be gained by finding out who is interested in funding that project.

For example, if the defence department offers contracts for research on a new type of synthetic chemical, scientists may infer that this chemical has military applications and therefore is likely to be subject to secrecy measures. Military or industrial secrecy is acceptable to some scientists, but not to others. The decision to work on research of great potential risk to humanity, such as nuclear weapons or germ warfare, is considered by many scientists to be a matter of personal ethics, not one of professional ethics.

Activity 3

- 3.1** Society uses the term 'chemical warfare' to refer to the intentional exposure of people to toxic substances that can kill or incapacitate them. These agents can cause people to collapse by lowering their blood pressure, putting them to sleep, causing severe discomfort or blocking the operation of their involuntary muscles.
- a** Do you think that the military should research and develop non-lethal physically incapacitating agents? Explain your opinion.
 - b** What if the military is only developing chemical agents so that they can find a cure for them, to help counteract any future chemical terrorist attacks? Should this be allowed?
- 3.2** Assume that an incapacitating agent that can immobilise people has been developed.
- a** Under what conditions should it be used?
 - b** Should it be used on a battle field, for example, if Australian soldiers are being attacked?
 - c** Should it be used against rioting prisoners in order to regain control?
 - d** Should it be used when capturing suspected terrorists?
 - e** Should it be sold in 'pocket packs' for individuals to use for personal protection?
 - f** Should it be given to children to carry in their pockets in case they are abducted by strangers?

For and against

Scientific discoveries are important to everyone – children, students, adults, rich people, poor people, those who live in the country and those who live in cities. Scientists and their discoveries have made possible the world we have today – the conveniences like television and cars and the achievements like space travel, as well as the problems like pollution and the use of weapons of mass destruction.

Activity 4

- 4.1** Do you think that scientists should be held responsible for the ways their research is applied? Or should they only be concerned with the scientific discoveries, and let other people decide how these discoveries should be applied? Some views about this are listed below. Decide whether you agree or disagree with each one, and explain your answers.
- a** Scientists should have some responsibility for the ways their research is used, as they know the details the best.
 - b** The sponsors who provide the funding for the research should decide how the discoveries will be used.
 - c** If the products or discoveries could cause environmental or social damage then the scientist has an obligation to stop anyone from using their work.
 - d** It is the scientist's research, so the scientist should have complete responsibility for the applications of their work.
 - e** Once a discovery is made, the scientist's work is complete. If someone else uses their work irresponsibly or to cause harm, then it is that person's responsibility, not the scientist's.



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

Science changing behaviour

To deal with many 'big issues' such as disease prevention and climate change, people have to change their behaviour and governments have to change their policies. In order for people to choose to make these changes, we need to understand how to access evidence, consider it and make rational conclusions from it. In other words, we need to use a scientific way of thinking to help us to identify and discard any prejudice involved in an issue or problem.

This way, people can make the best possible decisions based on the valid evidence.

Real decisions from real science

Making choices is about decision making. It involves the mental process of judging the qualities of more than one option, and selecting the best option. The important part is recognising that there is a choice to be made. It is also important to consider the possible consequences of our choices.

Activity 1

1.1 Consider these four typical responses of people to decision making.

- ▲ You want to know everything about an issue before making a decision.
- ▲ You accept what you read in the media and go with the flow.
- ▲ You avoid making choices until you really have to.
- ▲ You have trouble coping with uncertainty and the possible consequences and delay making a decision until it is too late.

How do you react to difficult decisions?

Activity 2

2.1 The details below of a recent newspaper report warn about a link between artificial sugar in foods and cancers.

The report says that researchers have strengthened a link between aspartame – a common artificial sweetener in foods – with cancer in rats. It goes on to explain that the research was carried on by a team of scientists working for a reputable, independent, non-profit national research foundation. The results of their research were published in the government-sponsored, peer-reviewed journal *Environmental Health and Food Safety*.

The executive director of a well known public 'watchdog' group believes that the study raises serious questions about the safety of using aspartame in foods, and she is calling for a re-evaluation of the chemical.

Dr Sugar, a dietitian who owns a calorie-control company, maintains that the methodology was faulty and its results would unnecessarily alarm people. She also believes that the design and execution of the investigation did not follow the government guidelines set down for this type of research.

The article quotes the lead researcher as saying 'the incidence of cancers was significantly higher in rats exposed to the sweetener than in rats not exposed to the sweetener ... based on our scientific data, we believe that aspartame should be avoided as much as possible by pregnant women and children.' He also stressed that this is not just his opinion.

The report states that the acceptable level of aspartame intake per kilogram of body weight is 40 mL/kg. It also states that this is equivalent to about six cans of diet soft drink per day for a 23 kilogram child.

Aspartame is also found in yoghurts, sugar-free desserts, chewing gum and some medicines, so people's intake could actually be much higher than they think. The lead researcher has called for immediate government overhaul of regulations governing the use of aspartame as an artificial sweetener in foods.

A scientist from a different country who is in the same line of research maintains that the study did not follow the national guidelines and that the methodology and results are therefore faulty.

- a** Why is it important to know that this research was done by a team of scientists working for a reputable, independent, non-profit, national research foundation?
- b** What is aspartame?
- c** Is it ethical to deliberately cause malignant tumours to grow in rats?
- d** What are some of the problems with using rats to simulate situations in humans?
- e** Where was this research published?
- f** Is this a reputable scientific journal? Explain your answer.
- g** As a result of this research, who is raising questions about the safety of aspartame in foods?
- h** Why would a public 'watchdog' group become involved in this report?
- i** Who is claiming that the methodology was faulty?
- j** What personal interest does Dr Sugar have in raising questions about the validity of the research?
- k** Do you think people will believe Dr Sugar's criticisms of the research? Why or why not?
- l** Given that six cans of diet soft drink per day is the limit for a 23 kilogram child, estimate how many cans would be the limit for a 79 kilogram adult.
- m** What did the lead researcher mean when he said that this is not just his 'opinion'?
- n** Why do you think the scientist from a different country also discredited the research?
- o** After considering only the information provided by this report, what is your opinion about the use of aspartame in foods?
- p** What further evidence would you need to consider in order to make a fully informed decision?



Now check out the PowerPoint slideshow and the activity sheet on the CD-ROM.

1 – Is seeing believing?

1.1

Sense	Job
Eye	Detects light
Nose	Detects chemicals in the air
Tongue	Detects chemicals in substances in the mouth
Ear	Detects vibrations in the air
Skin	Detects temperature, touch, pain

- 1.2 **a** Sight **b** Hearing **c** Touch and sight
 2.1 **a** A tree with seedlings growing under it. **b** The seedlings are mostly on one side of the tree. **c** For example, there may be more nutrients or sunlight on this side.
 4.1 **a** The second **b** The second
 4.2 **a** Sight **b** Sight **c** Touch
 5.1 **a**
 5.2 No – this is an inference.

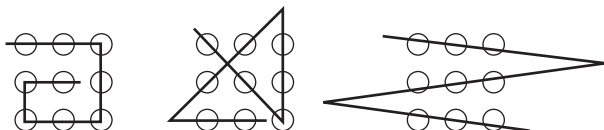
2 – Give it a name!

- 2.1 **b** Three **c** Two **d** No **e** Probably both
 2.2 Rattles – a, b, c, d, e, f. Crocodile subgroup – a, c, d, f. Snake subgroup – b, e.
 3.1 **a** Venus fly trap **b** Muscipula **c** Dionaea **d** Binomial system
 4.1 **a** Purple, it's a plant and it is carnivorous. **b** Carnivorous **c** 'Big teeth' **d** Carnivorous plants

3 – Dreaming up solutions

- 1.1 **a** Pour the water from glass number 4 into glass number 1 and return it to its original position. **b** Realising that the water could be poured and not depending on rearranging the actual glasses.
 1.2 **a** Three, as once you have chosen two the next one will be either a red or a green.
 2.2 Because he couldn't imagine how much more efficient the telephone would be, compared to messenger boys.
 2.3 Probably reduced the amount of work they had, and could have put them out of business.
 2.4 **a** Sample answers: living in space will be common, smaller and cleaner technologies for medicine, alternative energies such as fusion, virtual classrooms, microjets used for air travel on Earth instead of planes, solar smart cars, time travel to other planets, cloning. **b** Sample answers: human nature, being happy and sad, enjoying recreational activities, art and entertainment.
 3.1 **a** The residual air kept causing the filament to burn. **b** Pumped a gas into the bulb instead of trying to remove the air. **c** One of the inert gases, as they do not support combustion.

5.1



- 5.2 **a** They are all scientists of note except Gilcrest who is a cricket player. **b** The initial letters of the first seven names spell SCIENCE.
 5.3 Stand the signpost up so the arm indicating the place she has come from is pointing in the same direction. The other arms will then all point in the correct directions as well.
 5.4 It contains all the prime numbers and they are all in alphabetical order.

4 – Design – from concept to construction

- 1.1 **a** Polymer A is soft, flows easily, recovers quickly after impact and provides bounce. **b** Polymer C is hard, resists flow, does not recover quickly after impact, it is strong and provides some bounce. **c** This type of testing screens materials and products to determine whether a particular material meets the required standards.

- 2.1 **a** Polymer A **b** Viscous plastics absorb impact forces during the initial ground contact and provide a 'cushioning' effect, so in the rearsole to protect the heel. **c** In the midsole under the flexing part of the foot. **d** Polymer B, as it has some of both qualities.
 2.2 Foresole – polymer B. Midsole – polymer A. Rearsole – polymer C. Comments should show some degree of logic.

5 – What is a good hypothesis?

- 1.1 **a**
 1.2 Sample answers: **a** The sound of burning trees travels faster at night. **b** Native animals can smell smoke before humans. **c** Burning trees produce more heat than burning grass.
 2.1 The respiration rate of goldfish will decrease when the temperature of the water is lowered.
 2.2 Lions are mostly found resting in umbrella thorn trees during the heat of the day.
 2.3 **a** Long whales breach less often: Not supported by the observations.
 Thin whales breach less often than round whales: True, but has no connection to communication.
 Breaching whales flick their tails when they re-enter the water: Irrelevant to the hypothesis.
 Whales that breach most often live in groups: Correct, as those that travel in groups have more need to communicate.
b Round whales need more air than thin whales: Most likely to be correct.
 Whales breach so that they can dive deeper: Not supported by the observations even though round whales do breach more often than thin whales and as they breathe air while breaching.

6 – Determining the variables

- 1.1 **a** KrakFree Sunglasses do not break when dropped. **b** The type of surface they were dropped onto **c** Whether they broke or not **d** The height of the drop
 2.1 **a** Nitrate levels in the soil **b** Plant growth
 2.2 **a** Group with no fertiliser added **b** Group with fertiliser added **c** Because if the dependent variable changes in the experimental group, but not in the control group, the change can only be attributed to the independent variable.
 2.3 Ambient temperature, amount of sunlight, rainfall, other mineral levels in the soil, plant variety, seeds – use ones of the same age and from the same district.
 2.4 All the variables listed in the previous answers.
 2.5 Same factors as listed in the answer to 2.3.
 2.6 Growth of the plants with the fertiliser must be more than the growth of the plants in the control group.
 2.7 Revise the hypothesis or check the design of the experiment.
 3.1 **a** Positive **b** Positive **c** Positive **d** Negative

7 – Great experimental design

- 1.1 **a** 8 **b** 1 **c** 7 and 6
 1.2 **b**
 1.3 Weather conditions
 1.4 Age
 2.1 **d**
 2.2 **d**
 2.3 **c**
 2.4 **d**

8 – Measuring up

- 1.1 **a** 55 mL **b** 3.4 V **c** 10 mL, 40 mL
 1.2 0.68 V
 1.3 30 mL
 2.1 Measurement 2 (268 g) – measurement 1 (210 g) = mass (58 g). Increase in mass = 4 grams
 3.1 **a** 80 cubic centimetres **b** Cubic centimetres
 3.2 0.30 cubic metres
 3.3 **a** Spring balance **b** Newtons (N) **c** 5 N **d** 8 N

- 4.1 Human eye, magnifying glass, light microscope, electron microscope
 4.2 **a** In order to see the image, light must pass through the specimen.
b Resolving power (sharp images) is better in electron microscopes, but is limited to molecules made of about 300 atoms. It also has almost unlimited depth of field.
 4.3 **a** Magnification **b** Resolution **c** Field of view
 4.4 16

9 – Number crunching

- 1.1 **a** 2 **b** Time to reach 200 km/h **c** Time: seconds; weight: tonnes
 1.2 No table title, independent and dependent variables in wrong column, heading for each column is not detailed enough, no units supplied.
 2.1 Codes (SS) in table order: E, F, C, B, G.
 Codes (SD) in table order: B, G, E, F, C.
 3.1 **a** Vinegar test **b** Sugar test **c** Iodine test **d** Acid–base test
e Observation test **f** Water test
 3.2 Bases: toothpaste, bleach. Acids: shampoo, dish detergent, lemon juice

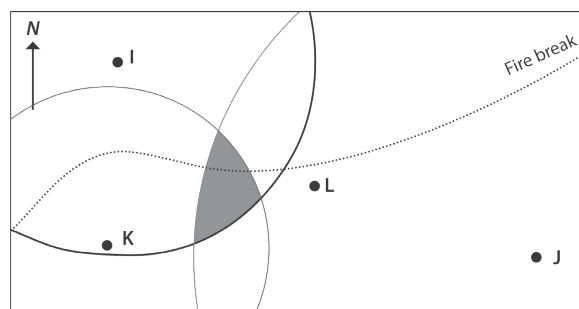
10 – In the wild – the ultimate laboratory

1.1 **a**

Head	Large casque on top of its head	No casque on top of its head
Legs	Short thick legs	Longer, thinner legs

b 24 cm

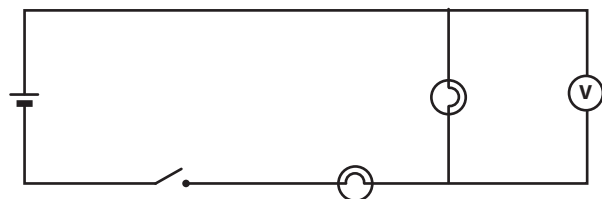
- 1.2 **a** The most recent fire in an area would affect the vegetation composition more obviously than the fire that occurred in 1998. 'L' is in the lower section of the area shaded with triangles, an area that a fire passed through 4 years ago. Using the graph, at 4 years after the fire there would be about 22% of the ground covered by shrubs and about 60% covered with grass. There will, however, be no trees.
b 2007
 1.3 There will be about 25 animals at location 'T' in the national park by 2008. Four years will have gone by without a fire through 'T'. The graph shows location 'L' had 25 animals after 4 years without a fire, and so it could be expected that 'T' will have 25 animals after 4 years without a fire.
 1.4 **a** J: 9 km, K: 4 km
b/c



1.5 Further than about 20 km

11 – A picture tells a thousand words

1.1



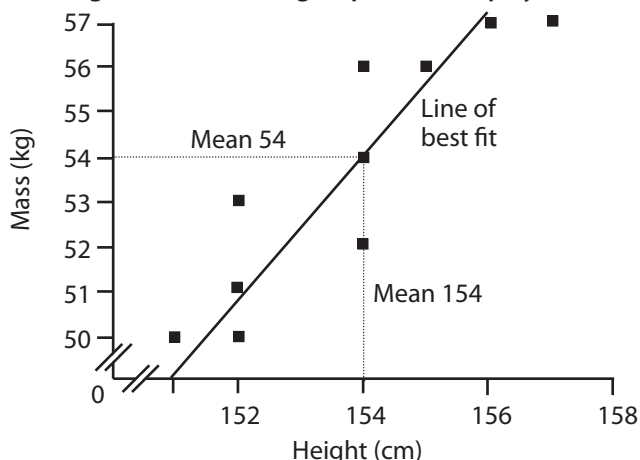
- 1.2 Diagram A
 2.1 **a** Temperature in degrees centigrade **b** Wind speed in knots
 2.2 **a** 7–10 knots **b** They were more exposed to the wind at the top, as there was less vegetation. **c** Zero **d** 'Exposed skin freezes within one minute.'

- 3.1 **a** The lines show how light refracts. **b** The white light source
c Red
 3.2 **a** The path of light through the lens **b** The object **c** The image of the object
 4.1 **a** 5.4 mm **b** A
 4.2 **a** 10 times **b** 8.6 mm

12 – Integrating information

1.1 **a**

Height and mass of a group of netball players



b Mean height = 154 cm, mean mass = 54 kg

- 2.1 **a** Yes **b** 2 h: 0.005 g/L. 4 h: 0.011 g/L. 6 h: 0.005 g/L. Every four hours the difference is the same. **c** They tend to converge and diverge every 4 hours.
 2.2 **a** About 160 mg/L **b** 40 days **c** Algae numbers increase until oxygen levels fall to about 75 mg/L and then they start to die.
d Oxygen levels fall faster.
 3.1 **a** 39 cells/mm³ **b** 210 mg/g
 3.2 **a** Line B **b** Between a and d
 3.3 **a** Between a and b **b** Between d and e **c** Algae growth is slowing down, reaching its peak and beginning to fall. **d** Algae numbers increase until available oxygen levels begin to affect the rate of growth, then the numbers begin to fall as algae starts to die.
 3.4 **a** 60 cells/mm³ **b** 26 mg/g oxygen

13 – What does it all mean?

- 1.1 **a** 6 **b** Glass **c** Air-filled plastic, 31 minutes **d** 3 minutes
 1.2 Glass, nylon, leather, aluminum, polystyrene, air-filled plastic
 1.3 Air-filled plastic
 1.4 Because it involves a layer of air
 1.5 **a** Still air is the best insulator. **b** It is expanded plastic so also contains air.
 1.6 Set up several cans in each insulator and average the results. Repeat the whole experiment several times.
 1.7 Better control of variables like uniform thickness of the containers, uniform space between the container and the can, the exact starting temperature of each can. Check that temperature was read accurately and consistently.
 1.8 Use other materials for the containers, test the insulators for use in eskies, try keeping other types of drinks cold, conduct more detailed work on heat transfer in materials in general.
 2.1 **a** For a line graph with a straight line through the origin, the conclusion may be that the extension is proportional to the load up to its elastic limit.
b The strap extends by 0.33 cm per Newton.
c The line indicates that there is a constant rate of change and that it is positive. There is an increase in both variables so they are directly proportional.

14 – Learning from each other

- 1.1 **a** An hypothesis is an educated guess, a conclusion is what was found out.
b A conclusion is a statement summarising what you found out. An inference is a suggestion based on observation.
c The results display the data, the discussion reports on the analysis of the data.
d The introduction is a brief section that orients the reader to the topic of investigation. The literature review is a summary of what is already known about the topic.
e The method section outlines the steps to be taken and the equipment section lists the things needed to carry out the investigation.
- 1.2 **a** O **b** O **c** C **d** H **e** O **f** O **g** C **h** H
- 2.1 **a**
- 2.2 **b**
- 2.3 The higher the sign is moved, the slower the truck moves.

15 – It's all about common sense

- 2.1 **a** Harmful **b** Corrosive **c** Explosive **d** Toxic **e** Radioactive
f Flammable
- 3.1 **a** Flammable: substance will burn given the appropriate conditions. Toxic: poisonous. Corrosive: substances that destroy living tissue. **b** No **c** Gloves, safety goggles, laboratory coat, ventilation
- 3.2 **a** May explode in the right conditions. **b** Don't eat it. **c** Do not breathe in gases produced. **d** Do not get any of it in your eyes. **e** Poisonous if absorbed through the skin.
- 4.1 **a** Water **b** Do not use a fire extinguisher – use sand in an emergency. **c** Carbon dioxide or dry chemical **d** Vaporising liquid

16 – Understanding models

- 1.1 **a** Will the genetic defect 'die out' with enough generations?
b What happens to a dummy in a head-on car crash?
c What would the atmosphere be like if its molecules were to act like tiny, speeding marshmallows instead of tiny, speeding hard balls?
d What will happen if the angle of the Earth's axis is changed?
e Will the new drug cure the disease?

2.1

Performance of a model boat in a tank	e
Hastening the investigation by using larger doses of drugs.	h
The destructive testing of products – using machines to simulate the wear on shoes.	g
Using a model of water flowing through a hose to describe electrons flowing through a circuit.	i
Computer-produced predictions of such phenomena as global weather, El Nino, volcanoes, tidal waves etc.	d
Rats may be sensitive to a drug that people are not.	a
Complex models of such things as global population growth, weather and global food supply	c
Recording the very rapid motion of birds and dancers.	b
Giving massive doses of chemicals to laboratory animals to try to get in a short time, the effect that smaller doses would produce over a long time	f

- 2.3 **a** The relative positions of the orbits of the planets in our solar system.
b This model does not show accurate distances or sizes, the 3-D nature of the orbiting planets, or the actual shapes of the orbits.

17 – Scientific attitudes

- 1.1 That continuity and stability are just as characteristic of scientific knowledge as change is.
- 1.2 No – scientists believe that science is a human discipline, therefore cannot be perfect.
- 1.3 No – science is a modification of ideas rather than their total rejection or acceptance.

- 1.4 By studying one part of the universe, the information gained can be applied to other parts.
- 1.5 **a** Law **b** Proof **c** Theory **d** Law **e** Theory
- 1.6 [In order] knowledge, observation, theory, new, challenge, theories, truth, law, true
- 2.1 **a** Kepler
b The force of gravity between any two objects in the universe depends only on their masses and the distance between them.
c The tides, the orbits of planets and moons, the motion of falling objects and the Earth's equatorial bulge
d Galileo's theory of relativity
e Philosophers outside science were concerned that if all motion runs according to mathematics, then the human notion of 'free will' may be only an illusion.

18 – Are you scientifically aware?

- 1.1 There will be a range of opinion but there should be evidence of consideration of each issue.

2.1

Headline	Is that likely to be so?	Is it really so?
Space tourism	Yes	Not yet but people have paid to be taken into space already
From the sewer to your glass!	Yes	It may well become so in Australia
First cloned human born	No	No

3.3

Headline	How did some people react?
a	Stopped using aluminium cookware
b	Live in fear of imminent environmental disaster
c	Stopped eating margarine
d	Started stockpiling gold in case of a global economic collapse
e	Adopted paranoid behaviour about sunlight

20 – Don't believe everything you hear!

- 2.1 **a** Increases the credibility of the results
b It's an artificial sweetener used mostly in low calorie diet foods.
c Opinion will vary.
d Rats may react quite differently to aspartame compared to people.
e Environmental Health and Food Safety
f Yes, because it is a government sponsored, peer-reviewed journal.
g A public 'watchdog' group and the research team itself.
h It is the role of this type of group to scrutinise information and provide constructive criticism.
i Dr Sugar, the owner of a calorie-control company, and an independent scientist in another country.
j Dr Sugar has a financial interest, as aspartame is an artificial sweetener and is used extensively in weight-control foods which is how his company earns its money.
k Dr Sugar is a doctor and a dietician and so people will probably take notice of his opinions even though they could be biased.
l 8 cans
m He is convinced that the result he is claiming is supported by the data.
n This is the process of peer review valued by all scientists. However, he may also be close to proving this link himself and want to be the first to publish.
o Opinion will vary according to individual analysis of the information.

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